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БЮДЖЕТНОМ ОБРАЗОВАТЕЛЬНОМ УЧРЕЖДЕНИИ ВЫСШЕГО
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ПАТРИСА ЛУМУМБЫ» ИНСТИТУТ ЭКОЛОГИИ**

As a Manuscript

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**ASSESSING THE IMPACTS OF URBANIZATION ON PLANT DIVERSITY AND
ECOSYSTEM FUNCTIONING IN TROPICAL CITIES: A CASE STUDY OF BENUE
STATE, NIGERIA**

1.5.15 ECOLOGY

**DISSERTATION FOR THE DEGREE OF
DOCTOR OF BIOLOGICAL SCIENCES**

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TO WHOM IT MAY CONCERN

Subject: Confirmation of Data Provision for Academic Research

This is to certify that Shualbu Ochoche a researcher from Institute of Environmental Engineering, People's friendship University Russia named after practice Lumumba (RUDN University), MOSCOW, Russia, Was provided plant-related data covering Makurdi local government area by our organization (Joseph Sarwuan Tarka University, Makurdi – Benue state-Nigeria) on request. The data was officially released to support his ongoing academic research and analysis. It includes records and information on (Brief description of the types of data e.g: plant distribution, species list, ecologies zones, etc), and was shared solely for academic and research purposes. We acknowledge that data originated from the records and provided directly to the above named researcher.

For further inquiries and verification, please feel free to contact our office

Sincerely,

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INTRODUCTION

Relevance of the Study. Urbanization can be seen as the main cause of ecological change in tropical regions, but its impacts on biodiversity and ecosystem functioning are still understudied in many rapidly growing African cities. Understanding how expanding built-up areas alter plant communities, soil processes, and vegetation productivity is important to detect long-term environmental consequences [1]. In recent studies [1][2][3], it is evident that tropical cities in particular face intense pressures as a result of high population growth, weak land-use regulation, and high dependence on natural resources, making them global hotspots for biodiversity decline and habitat fragmentation. To address this gap, this study provides spatial, ecological, and statistical evidence of how urbanization affects plant diversity and ecosystem functioning in Benue State, Nigeria.

This study is particularly relevant because it brings together remote sensing, field-based biodiversity surveys, and soil ecosystem indicators an integrated approach that is still rare in West African ecological research. Most previous studies have focused either on satellite-derived vegetation analysis or on field species surveys, often treating them separately and rarely examining urban, peri-urban, and rural areas together [4][5]. To address this gap, this study adopts a more holistic view of ecological health, capturing how land-use change interacts with vegetation structure and soil nutrient dynamics [6].

This study is also important for urban planning and environmental policy. Many Nigerian cities, including Makurdi and Otukpo, lack up-to-date ecological information to guide land-use decisions, green-space protection, and climate adaptation efforts. In their respective studies, [6][7] shows that integrating biodiversity indicators and vegetation metrics such as NDVI into planning processes can greatly enhance urban sustainability. The outcome of this research will provide policymakers with practical evidence-based guidance to develop green infrastructure, conserve remaining natural habitats, and support ecosystem services that are essential to human well-being.

Finally, this study adds to global discussions on how urbanization influences ecological processes, especially in data-limited regions. By developing a statistical model that links the intensity of urbanization measured through built-up density, night lights, and population density with ecosystem indicators such as NDVI, organic carbon of the soil, and nitrogen, the research improves understanding of urban ecological gradients in tropical settings. Again, for researchers studying similar urban transitions in Sub-Saharan Africa and other rapidly urbanizing tropical

regions, the study finding will provide them with useful comparisons [8][9]. In doing so, the study not only addresses regional data gaps, but also strengthens broader ecological research and practice.

The Degree of Development of the Problem:

Although urban ecological issues are receiving increasing global attention, the past literature on how urbanization affects plant diversity and ecosystem functioning in tropical African cities remains relatively scarce. Existing studies have largely concentrated on temperate regions, where long-term ecological monitoring systems and high-resolution datasets are more readily available [11]. In contrast, cities in tropical Africa, particularly in West Africa, are experiencing rapid and often poorly regulated land-use changes, yet are much less represented in the scientific literature. The general knowledge gaps are continued with respect to the impacts of accelerated urban growth on local plant communities, soil conditions, and vegetation productivity along rural–urban gradients [10][11].

There is rare literature that explicitly connects these parts: urbanization, biodiversity, and ecosystem functioning in Nigeria. Most existing studies tend to focus on separate components such as LULC, plant diversity, or soil degradation without examining their combined or interactive effects within an integrated framework [10]. However, some of the literature that applies quantitative urbanization indicators, including night-time light intensity, built-up density, and population concentration, remains particularly limited, despite growing evidence that these metrics are strong predictors of ecological stress on a global scale [12]. This fragmented approach has constrained a holistic understanding of how urban expansion is reshaping ecological processes in rapidly growing cities such as Makurdi and Otukpo.

The problem is also underdeveloped in terms of methodological integration. Although remote sensing technologies have improved substantially, their application to tropical urban ecology is still limited by insufficient ground-truth and the absence of fine-scale biodiversity data [15]. Similarly, in Nigeria, the literature focused on plant diversity is often short-term and restricted to small sampling areas, lacking the extensive spatial stratification required to capture urban gradients [16]. In addition, Soil indicators such as organic carbon and nitrogen that are vital in assessing ecosystem functioning are rarely incorporated into ecological impact assessments despite mounting evidence of their sensitivity to urban disturbances [17].).

The conceptual and empirical understanding of this issue remains incomplete. Because, only a limited number of studies have applied statistical modelling approaches to quantify how multiple dimensions of urbanization jointly influence vegetation health, soil conditions, and the structure of the plant community in tropical settings. The most recent innovation in the field of ecology emphasizes the pressing need for integrating data-driven frameworks that combine remote sensing, species inventories, and ecological indicators to better capture the complex impacts of urbanization [13][14].). By responding to these gaps, the present study makes a meaningful contribution to advancing research on this largely understudied problem in tropical cities—particularly within Benue State and Nigeria in general.

Understanding the impacts of urbanization and its challenges on natural ecosystems is important. The role it plays in shaping the ecological and economic sphere in the Benue state and in Nigeria at large has led to different research by several scientific studies (Shabu.T, Garba U., Job I.o, Mala.M, Mabogunje A.L, Onokerhoraye A.G, Eghaosa E.O, Ismaila.T, Gue ssan K.N., essan.K.N, Jelili.M.O, Ajibade A., Alabi A.T, Stren R.E, Faniran.A. Onanuga, M. Y., Nnaji, C. C, Tyubee, B. T., Onilude, O, Ogunbode, T.

Objects of Research

We study and assess the effects of urbanization on plant diversity and ecosystem services in Makurdi and Otukpo, two rapidly urbanizing cities in Benue State, Nigeria. This study examines LULC over three decades ranging from 1990-2024 to understand the impact of urban expansion on local plants.

Subject of Research

Multi-temporal satellite imagery and ancillary geospatial datasets were used that are necessary to detect land-use/land-cover transformations, quantify vegetation condition, and assess urbanization intensity in Makurdi and Otukpo. Landsat TM (1990), ETM+ (2010) and OLI (2024) scenes were obtained for change detection, NDVI, and EVI derivation due to their 30 m spatial resolution and long-

term continuity, which makes them suitable for landscape-scale ecological studies. Downloading Landsat imagery that covers both Makurdi and Otukpo, at interval of 10 years between 1990, 2010, and 2024 years for possibly three periods. VIIRS Day/Night Band night-time lights imagery was used to represent anthropogenic intensity and built-up expansion through illumination patterns, while WorldPop population density rasters provided high-resolution demographic information to quantify population-driven urbanization pressures, these were extracted at the ward level.

Aim OF The Study

The main objective and purpose of this study is to investigate the impacts of urbanization on plant diversity and ecosystem functioning in Benue State, with a focus on changes in LULC patterns and the resulting effects on ecosystem services and plant biodiversity. The selected time periods for analysis are 1990, 2010, and 2024, reflecting significant phases of urbanization and environmental change in the study area.

The Purpose of the Dissertation

Assess how urbanization influences plant diversity and ecosystem functioning across urbanization gradients in Makurdi and Otukpo LGAs Benue state, using an integrated approach combining remote sensing, field-based biodiversity assessment, soil indicators, and statistical modelling.

Analyze the changes in LULC in Makurdi and Otukpo for 1990, 2010, and 2024, and quantify the trends and extent of urban expansion and vegetation loss across rural, peri-urban, and urban zones.

Assess plant species richness, diversity, and evenness across urban gradients using field-based vegetation inventory and compute indices such as Shannon index, Simpson index, and Pielou's evenness.

Evaluate how increasing urbanization influences the functioning of the ecosystem in rural urban gradients and statistically determine whether urbanization significantly affects plant diversity and soil ecosystem quality.

To develop and propose evidence-based policy recommendations for sustainable urban planning and biodiversity conservation in tropical cities, based on the study's findings on the impacts of urbanization on plants and ecosystem function.

Scientific Novelty.

- The systematic sampling of 33 vegetation plots across all wards in Makurdi and Otukpo further provides one of the few fine-scale biodiversity datasets for the region, strengthening empirical understanding of species-environment interactions in tropical cities.
- The study contributes to global scientific debates on how urbanization shapes ecological processes, particularly in regions where data scarcity limits robust analysis.
- This study integrates field-based plant diversity assessments, soil ecosystem indicators, and remote sensing metrics to evaluate urbanization impacts in a tropical city, a combination rarely applied in Nigerian or wider tropical urban ecological research.
- Through a quantitative regression framework, the research introduces a more robust, multi-dimensional assessment of anthropogenic pressure, an approach underutilized in ecological studies in the research area, by Connecting NDVI to built-up density, night-time lights, and population density
- The methodological and empirical contributions of the study advance current knowledge on how rapid tropical urbanization shapes plant diversity and ecosystem functioning, addressing well-recognized research gaps in the urban ecology literature.

Tasks of Dissertation

Remote sensing served as the foundation for quantifying land-use transitions for a selected period from 1990, 2010, 2024, and vegetation dynamics, using satellite-derived indices and classification products processed within the Google Earth Engine (GEE) environment. The Field based plants sampling was carried out across 33 stratified plots, representing urbanization gradients in the 6 wards of Makurdi and 5 wards of Otukpo (three plots per ward. For each plot, GPS coordinates, elevation, land-use context, and additional plot metadata were documented to ensure accurate geolocation and integration with remotely sensed variables. These data supported the computation of biodiversity indices (Shannon, Simpson, Evenness) and enabled the comparison of species composition along the rural-peri-

urban–urban gradient. Complementary to the plant survey, geo-coordinates of the plots were recorded.

Statistical analysis, including ANOVA tests to compare vegetation and soil parameters along the urbanization gradient, with assumption checks for data distribution and homogeneity. The Regression modelling was implemented to assess the relationship between urbanization metrics and vegetation condition, using the specified model. The results of the vegetation indices showed the clearest ecological signal. NDVI declined from 0.482 ± 0.023 (rural) to 0.277 ± 0.076 (urban), while EVI decreased from 0.312 ± 0.025 to 0.173 ± 0.051 , demonstrating stark reductions in primary productivity and vegetation vigor. ANOVA results confirmed highly significant differences exist across the gradients zones (NDVI: $F = 31.04$, $p < 0.001$; EVI: $F = 32.52$, $p < 0.001$), revealing that vegetation functioning diminishes sharply with increasing urban intensity. The indicators of ecosystem functioning showed a marked decline with increasing urban intensity. The NDVI values have decreased from 0.48 in the rural areas to 0.28 in the urban areas, while the EVI values also have reduced from 0.31 to 0.17, which indicate the reduction of vegetation vigor and the primary productivity. However, the results of the soil analysis showed that there is a significant increase in soil pH in urban zones ($p = 0.037$), suggesting that there is an altered soil chemistry linked to anthropogenic disturbance. Although SOC and nitrogen appeared to be higher in urban zones, these increases are likely driven by waste inputs rather than natural fertility processes. However, regression modelling further showed that population density is the strongest suppressor of vegetation greenness having this value ($\beta = -0.0072$, $p < 0.001$), highlighting demographic pressure as a key driver of ecological degradation.

To accurately know if vegetation indices differ significantly between urban zones, one-way ANOVA tests were conducted for NDVI and EVI. The findings indicate highly significant differences where (NDVI: $F = 31.04$, $p < 0.001$, $\eta^2 = 0.75$; EVI: $F = 32.52$, $p < 0.001$, $\eta^2 = 0.76$), which shows the impact urbanization has on plant diversity.

The overall performance of the model ($R^2 = 0.862$) shows that NDVI variability is accounted for by these three indicators, underscoring the central role of urbanization dynamics in shaping ecological conditions. The strong negative effect of population density combined with the modest positive effect of built-up surfaces suggests that ecological degradation is predominantly related to demographic expansion rather than structural development alone.

Biodiversity Decline and Ecological Homogenization. The decline in species richness from 23.50 species (rural) to 15.83 species (urban) indicates that urbanisation fosters homogenisation of plant communities. Disturbance-tolerant species such as *Cleome viscosa*, *Sida acuta*, and *Heterotis rotundifolia* become more dominant, while native forest species decline. This shift reduces functional diversity, disrupts trophic and symbiotic relationships, and decreases ecological resilience.

Habitat Fragmentation and Landscape Conversion: The loss of 124.93 km² of forest and the widespread decline of croplands have fragmented formerly continuous habitats into isolated patches. Fragmentation reduces dispersal opportunities, increases edge effects, and disrupts plant community dynamics.

Theoretical and Practical Significance of the Work.

From a theoretical perspective, the study deepens understanding of the effect of urbanization on ecological patterns in tropical cities (urban ecology, landscape ecology, and socio-environmental systems). It advances existing theory by empirically connecting key urbanization measures such as built-up density, night-time light intensity, and population density with indicators of ecosystem functioning, including vegetation greenness (NDVI), soil nutrients, and plant diversity. The study demonstrated how urban growth and ecosystem can be modelled [15][16][17]. The findings provide new insights into resilience, species distribution patterns, and the pathways for ecosystem processes [18][19].

From a practical point of view, the results provide actionable insights for urban planners, environmental managers, and policy makers in Benue State and other tropical regions experiencing rapid urban growth. The high-resolution spatial and field datasets produced, including vegetation surveys, soil indicators, and land-use/land-cover change maps, can be directly applied to environmental monitoring, impact assessments, and local climate adaptation planning. Lastly, the findings identified priority areas for conservation and ecological restoration, also proven in past studies [20][21], which enables targeted interventions that reduce biodiversity loss and the enhancement of ecosystem services in mostly growing and expanding cities [22][23].).

Methodology and Research Methods. This study was carried out between 2022 and 2025 in the two phases of the study. The research design follows an integrated methodological framework that was structured around the main study objectives and by combining geospatial analysis, ecological field measurements, and soil

assessments. Remote sensing served as the foundation for quantifying land-use transitions for a selected period from 1990, 2010, 2024, and vegetation dynamics, using satellite-derived indices and classification products processed within the Google Earth Engine (GEE) environment. Field based vegetation sampling complemented these spatial datasets by documenting species composition and diversity across gradients of urbanization. Soil properties essential for understanding ecosystem functioning, such as Soil Organic Carbon, Nitrogen, Bulk Density, pH, and Cation Exchange capacity, were extracted using GEE and urbanization indicators were also extracted from the same GEE. The methodology used provides a comprehensive, multi-dimensional approach for achieving the purpose of the study.

The Main Provisions to be Defended:

The study examines how urban expansion, vegetation transformation, and anthropogenic pressure over a 34-year period (1990-2024) shape ecological patterns in rural, peri-urban and urban zones. The principal scientific provisions for defence in this work are summarized as follows:

Regression Analysis of Urbanization Metrics as Predictors of NDVI.

The model performed strongly, indicating that a considerable proportion of the spatial variation in NDVI is attributable to these three urbanization indicators. The overall model explained 86.2% of the variation in NDVI ($R^2 = 0.862$), with an adjusted R^2 of 0.842, confirming high explanatory power even after adjusting for the number of predictors. The model's F-statistic was significant where ($F = 41.73$, $p < 0.001$) indicating that the predictor set collectively contributes meaningfully to NDVI variation. Information criteria ($AIC = -80.28$; $BIC = -75.57$) also support a well-fitted model.

Synthesis of Findings Across Indicators

The vegetation indices showed the clearest ecological signal. The value (NDVI) decreased from 0.482 ± 0.023 to 0.277 ± 0.076 , while the EVI decreased from 0.312 ± 0.025 to 0.173 ± 0.051 , demonstrating significant reductions in primary productivity and vegetation vigor. ANOVA results confirmed highly significant differences between zones ($F = 31.04$, $p < 0.001$ in the case of NDVI; and $F = 32.52$,

$p < 0.001$ for the case of EVI), revealing that vegetation functioning decreases sharply with increasing urban intensity The ANOVA

Habitat Fragmentation and Landscape Conversion

The loss of 124.93 km² of forest land and also the drastic reduction in cropland size have fragmented formerly continuous habitats into isolated patches. This fragmentation, in turn, reduces dispersal opportunities, increases edge effects, and disrupts the dynamics of the plant community. And again, patch isolation also limits pollinator movement, seed flow, and successional processes essential for ecosystem recovery.

Biodiversity Decline and Ecological Homogenization

The decline in species richness from 23.50 species (rural) to 15.83 species (urban), along with the reduced Shannon and Simpson indices, indicates that urbanization promotes homogenization of plant communities. Disturbance tolerant species such as *Cleome viscosa*, *Sida acuta*, and *Heterotis rotundifolia* become more dominant, while native forest species decline. The shift reduced functional diversity, disrupted trophic and symbiotic relationships, and decreased ecological resilience.

Soil Fertility Modification and Biological Function Disruption

Significantly higher soil pH in urban areas ($p = 0.037$) indicates altered nutrient availability and potential micro-nutrient deficiencies that can suppress plant growth. Although SOC and nitrogen appear elevated, the biological reality is reduced nutrient turnover, poorer microbial activity, and compacted soil structure conditions that limit root development and reduce soil-plant interactions fundamental to ecosystem processes.

Reduced Primary Productivity and Vegetation Health

The substantial decrease in the value of NDVI from 0.48 to 0.28 and that of EVI from 0.31 to 0.17 tells of the weakened photosynthetic activity, reduced biomass accumulation, and lower canopy cover. As a result of the reduction, it compromises ecological functions such as carbon sequestration, microclimate regulation, soil stabilization, and hydrological buffering. Urban hot spots with the highest population densities show the lowest NDVI values, confirming that demographic stress is a major biological driver of ecosystem decline.

Statistical Models Confirm Urbanization as a Predictor of Ecological Degradation.

The regression modelling of these variables ($NDVI = \beta_0 + \beta_1 \text{ Built-up} + \beta_2 \text{ Night-time Lights} + \beta_3 \text{ Population Density} + \varepsilon$) revealed that the three indicators of urbanization had significantly predicted plant productivity in the study area. Random Forest regression also classified predictors by importance, confirming built-up area and night-time lights as the leading determinants of ecosystem functioning in both LGAs.

Evidence for Sustainable Urban Planning and Conservation Interventions.

The combined findings from the LULC change, biodiversity patterns, soil indicators, and statistical modelling support actionable recommendations for urban management in the Benue State. Key provisions include:

- Strengthening land-use zoning and control of built-up expansion.
- Restoration of degraded vegetation and promotion of urban forest.
- Protection of the remaining natural vegetation patches in peri-urban and rural zones.
- Integrating ecological datasets (NDVI, species diversity, soil metrics) into planning decisions.

These provisions align with global frameworks for nature-based solutions and provide scientific justification for policies that enhance biodiversity conservation and ecosystem resilience in tropical urban environments.

Degree of Reliability

All classified maps were validated with ground truth data collected from the field using the accuracy test tool in ArcGIS. A total of 11 wards were randomly selected (6 in Markurdi and 5 in Otukpo), where each ward had 3 plot samples and each plot sample covered an area of 100 m². All were surveyed in each zone. The plant species were manually identified and classified based on their life forms (trees, shrubs, and herbs) and growth forms (perennial or annual). The richness and abundance of the species were also recorded for each plot. This survey was carried out alongside the Department of Wildlife and Range Management, College of Forestry and Fisheries, Joseph Sarwuan Tarka University, Makurdi. All geospatial predictor variables used in this study including the enhanced vegetation index (EVI), the normalised difference vegetation index (NDVI), organic carbon, nitrogen, bulk

density, cation exchange capacity, pH, night-time lights, population density and built-up intensity were extracted using Google Earth Engine (GEE)[24][25][26][27].).

The workflow integrated multiple remotely detected and modelled datasets, and all layers were clipped to ward-level buffer boundaries, aggregated to mean values per ward, and exported directly from GEE into a structured table. The final datasets produced by this workflow is openly available in the Appendix, ensuring reproducibility as well as the national population data in appendix 10 were collected from the National Population Commission, Department of Population Census and Statistics, Headquarters, Abuja.

Compliance of the Dissertation with the Passport of the Scientific Specialty.

The dissertation work corresponds to the specialty passport 1.5.15. Ecology by item: Item 10. Anthropogenic impact on populations, communities, and ecosystems. Biological effects of environmental pollution with toxic substances (ecotoxicology). Development of biological methods and criteria for assessing the state of the environment, bioindication, biotesting, and biomonitoring. Development of environmentally sound standards for the impact of human economic activity on wildlife: item 12. Ecological principles of natural protection at the population-specific and ecosystem level.

Approbation of the work

The result of the dissertation research was the publication of 5 articles, including 3 in scientific journals indexed in the SCOPUS database and 2 in a journal included in the Higher Attestation Commission (HAC) list. The results of the dissertation work were presented at the following conferences: International youth scientific conference “EPI 2025”: Environmental studies and protection issues, Institute of Environmental Engineering RUDN University April 2025, XXVth International Multidisciplinary Scientific GeoConference Surveying, Geology and Mining, Ecology and Management (SGEM) Albena Resort & Spa, Bulgaria 28 June-7 July 2025, proceeding, Roles of Artificial intelligence in predictive Agroecological Modelling and sustainable land management in response to climate viability in Moscow Region (Scopus **A Pathway to Renewable Energy, Russia African Conference, July 2023** The Application of Remote Sensing Techniques in Monitoring and Assessment of Inorganic Pollutants in Africa, Anthropogenic

Activities: **A Threat to Natural Resources Sustainability Possible Solution** (Council of Young scientists of the institute of ecology of RUDN University **“Lecture Seminars on Studying the Earth” December 3, 3 & 10, 2022**), Urbanization Challenges and Its Pressure On Natural Resources Sustainability In Nigeria International youth scientific conference **“GREEN-2022” Graduate Research in Ecology, Engineering and Nature; November 2022 Institute of Environmental Engineering RUDN University**, International youth scientific conference “EPI 2024” Environmental studies and protection issues. Institute of Environmental Engineering RUDN University "Common Natural Ecosystem Challenges in Nigeria": **Assessment of urbanization's environmental impact in Makurdi Metropolis, Benue State, Nigeria. International Journal of Environmental Impacts**, Vol. 8, No. 2, pp. 393-400. <https://doi.org/10.18280/ije.080218> (Scopus Q3, A social approach to assessing the impact of urbanization on natural resources: A case study of Makurdi-Benue State, Nigeria) International Research Journal No6 (156). **URL: <https://doi.org/10.60797/IRJ.2025.156.82> Issue: No. 6 (156), 2025 (VAK K1(SCOPUS Q2, WOS Q1**, Spatiotemporal Analysis of Land Use and Land Cover Changes Driven by Urbanization in Makurdi, Nigeria (1990 to 2024): Implications for Cropland Loss, Environmental Risk, and Sustainable Urban Development (SCOPUS Q3 WOS Q3), Journal of Applied Environmental Research, Socio-ecological consequences of urban growth in dutsen kura gwari, Niger state, Nigeria, Soil fertility management practices among smallholder farmers in Ughelli North, Delta State, Nigeria. (2025), Tropical Agriculture, 102(3), 358–367.

Personal Contribution of the Author.

The candidate participated in setting the aim and specific objectives of the study, collecting and analyzing the material obtained, processing and interpreting the data, and prepared the publications in co-authorship.

Structure and Volume of the Dissertation. The thesis consists of an Introduction, 3 chapters, conclusions, Recommendations, List of Abbreviations, list of references, 11 appendices. The thesis is summarised and presented on 141 pages; the main text includes 15 figures and 21 tables.

Acknowledgement

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This chapter reviews the relevant literature that provides the scientific foundation for assessing how urbanization influences plant diversity and ecosystem functioning in tropical environments. It is organized around the main themes of the study, beginning with the concept, patterns and drivers of urbanization, followed by the documented effects of urban growth on plant species richness, evenness and community structure along rural-urban gradients. The chapter also examines ecosystem functioning metrics such as vegetation indices (NDVI/EVI), soil organic carbon, and soil nitrogen and their ecological significance, along with remote sensing and GIS techniques used in monitoring land-use change and vegetation dynamics. Furthermore, the review highlights statistical modelling approaches, including regression and machine-learning tools, that help quantify the relationships between urban intensity and ecological variables. The chapter concludes with a conceptual and theoretical framework that integrates these themes and guides the analytical direction of the present study.

CHAPTER ONE

1.1 Conceptual Clarifications

1.1.1 Concept of Urbanization

Urbanization is defined as the increase in the concentration of people, infrastructure, and economic activities in towns and cities, resulting in the physical and demographic expansion of urban areas. Historically, it evolved alongside industrialization in the past 18th and 19th centuries, as technological innovations, improved transportation, and factory-based economies drew populations from rural settlements into emerging metropolitan center's [34] <https://recentscientific.com/urbanization-concepts-dimensions-and-factors>. But in our contemporary era, urbanization has shifted from being an industrial phenomenon to a global socio-economic process shaped by globalization, service economies, and technological transformation. This shift is especially concentrated (pronounced) in the most developing nations of the world, where urban growth is characterized by unprecedented rates of human population increase, and, of course, it is often without corresponding improvements in infrastructure, environmental management, or governance [29][30]. Looking at the context of Africa and Asia, rapid urbanization is largely driven by natural population increase, rural-urban migration, expanding economic opportunities, informal settlements, and the physical extension of built-up land into peri-urban and agricultural zones[37]. In addition, the causes are majorly; the infrastructural development, political centralization, transportation corridors, and land speculation, all of which reinforce the conversion of natural landscapes into urban land uses [31][5]. The combination of these factors made urbanization (rapid urban growth) a powerful force to re-conform with which it is really shaping ecological patterns, resource availability, and environmental quality, particularly in low- and middle-income regions where urban planning systems struggle to keep pace with population and spatial expansion.

1.1.2 Urbanization in the Tropical Context

Urbanization in tropical regions is shaped by unique environmental, socio-economic and climatic conditions that distinguish tropical cities from those in temperate zones. The major characteristics of tropical cities are; high population densities, rapid spatial expansion, extensive informal settlements, insufficient infrastructure, and strong dependence on natural resources and ecosystem services [32][33].). The climate conditions of these cities are marked by high temperature, intense rainfall, and also a year-round biological productivity that increases sensitivity to land-use change, making vegetation cover, soil properties, and hydrological systems particularly vulnerable to urban disturbances. The evidence from the humid tropical regions is that the problem escalates to deforestation, altering of microclimates, reducing vegetation greenness, increasing surface runoff, and degrading soil fertility [41], which in turn leads to significant ecological consequences such as biodiversity loss, reduced carbon storage, and weakened ecosystem [34][35].).

Across Africa, and particularly in West Africa, urban growth has been exceptionally on the high and is driven by natural population increase, rural-urban migration, administrative centralization, and expanding economic activities in transportation and trade [43]. According to previous studies [36][37], West African mega cities such as Lagos, Accra, Bamako, and the emerging secondary ones such as Makurdi and Otukpo clearly exhibit similar patterns of fragmented expansion, infrastructural deficits, and strong interactions between human activities and ecological processes .

1.1.3 Land Use and Land Cover (LULC)

Land use refers to human purposes or activities applied to land such as agriculture, settlements, industry, transportation, and conservation reflecting socio-economic decisions and management practices imposed on land [46] and land cover describes the physical and biophysical features present on the surface of the Earth, including vegetation, water bodies, bare soil, built-up areas, and wetlands, regardless of how humans utilise it [47].). However, the two are conceptually distinct, but are closely linked [38]). According to Yaghoobi et al. 2022, it is vital to understand the linkages between human-induced land-use activities and the leading drivers of land-cover change globally, especially in rapidly urbanizing regions where natural landscapes are continuously converted into built-up surfaces [48].

Furthermore, from previous studies [39][40], accurate LULC mapping/classification can help environmental managers and researchers quantify and evaluate the key processes of vegetation loss, habitat fragmentation, soil degradation, hydrological disruption, and urban expansion that shape ecological responses at multiple spatial scales. To manage the cities of tropical and developing nations, where urban expansion is in a high level and is fast and often uncontrolled, LULC assessments are crucial to monitoring environmental impacts, planning green infrastructure, and implementing conservation strategies. Remote sensing and GIS on the other hand have become indispensable tools for proper and adequate LULC analysis, offering consistent, repeatable, and large-scale observations that support the evaluation of long-term ecological [41][42].). For studies like the present research, the dynamics of LULC provides the foundational spatial context to examine how urbanization gradients influence plant diversity, soil conditions, and vegetation productivity.

1.1.4 Gradient of urbanization

The urbanization gradient is referred to as a spatial continuum ranging from rural areas with low population density and minimal built infrastructure, through peri-urban areas which are characterized by a mixed land uses and rapid transformation, to highly developed urban centres dominated by impervious surfaces and intensive human activities [53].). This particular urban gradient framework is becoming more commonly used in ecological research [43][44][45], because it captures the progressive influence of urban development on environmental phenomena, therefore, allowing systematic comparisons of biodiversity, ecosystem functions, and habitat quality across varying levels of human engagements [56].).

1.1.5 Plant Diversity

Plant diversity comprises several complementary dimensions which are: 1) species richness, 2) the sheer count of different plant species, 3) species evenness-referring to how equally individuals are distributed among those species; and broader metrics such as alpha diversity (diversity within a site), beta diversity (turnover or change in species composition between sites), and gamma diversity (total diversity across a landscape) [46][47]; These dimensions offer different ecological insights and paths. For example, α -diversity helps understand the complexity of local communities, while β -diversity reveals how plant composition shifts across environmental or land-use classes [59]. High plant diversity plays a crucial role in the functioning and resilience of the ecosystem. Major empirical studies [48][49] have demonstrated that

greater species richness enhances carbon storage, boosts productivity, and improves multiple ecosystem services by promoting complementary resource management, stabilising community structure, and increasing resistance to environmental stressor/disturbances Kovacs et al. 2025 in their study of urban ecology, diverse plant communities support services such as air purification, microclimate regulation, pollinator habitats, and soil stabilisation, making plant diversity a key component of a sustainable and resilient city.

1.1.6 Ecosystem Function

Ecosystem functioning refers to the sum of ecological processes that regulate how ecosystems operate, including primary productivity, nutrient cycling, energy flow, soil formation, and vegetation [62]. These processes determine the capacity of ecosystems to sustain life and maintain stability, even if they are strongly influenced by parameters such as soil organic matter, microbial activity, vegetation cover, and hydrological balance [63].). In terrestrial landscapes, indicators such as net primary productivity (NPP), soil nutrient availability, infiltration capacity, and biomass accumulation are commonly used to assess ecosystem functioning, especially in environments undergoing rapid land-use changes [64]. A substantial collection of evidence has proven beyond doubt that biodiversity, particularly the richness and functional diversity of plant species, enhances the functioning of the ecosystem by promoting the use of complementary resources, stabilising productivity, and improving the retention of nutrients and soil [4][50].).). The biodiversity ecosystem functioning (BEF) relationship is critical in most urbanised landscapes of tropical regions of the world, where habitat fragmentation and vegetation loss can impair carbon sequestration, nutrient cycling, and soil all at once [66]), making plant diversity a key driver of ecological integrity across rural-urban gradients.

1.1.7 Vegetation Indices (NDVI, EVI)

Vegetation indices are quantitative measures derived from satellite spectral reflectance, particularly the differential reflection of red and near-infrared (NIR) wavelengths by vegetation available all in Landsat and Sentinel imageries, which allow researchers and environmental managers to assess plant vigorously and photosynthetic activity [67]. NDVI basically operates on the principle that any healthy plant absorbs red light for photosynthesis and strongly reflecting NIR radiation, producing values that correlate highly with biomass, productivity, and canopy [68]). EVI, on the other hand, builds on the limitations of NDVI by

incorporating blue band reflectance and correction factors to minimise atmospheric interference and canopy background noise, making it more highly and reliable in humid or densely vegetated tropical [69]. Both have become essential tools for evaluating plant ecosystem health, detecting vegetation stress, monitoring urbanisation impacts, and quantifying changes in ecological functioning across landscapes, especially in rapidly transforming tropical cities where field data alone are insufficient [51][52].). Their integration within a same GIS and remote sensing framework enhances the capacity to track reliable and efficient spatio-temporal vegetation patterns, providing robust indicators of ecological degradation, fragmentation, and resilience along urbanisation gradients for accurately and acceptable results suitable for policy recommendations [72].).

1.1.8 Soil Ecosystem Indicators

The properties of the soil ecosystem, especially (soil organic carbon (SOC), nitrogen (N), bulk density, pH, and electrical conductivity (EC)), are considered the fundamental measures of soil health and ecological functioning of any soil [73].). SOC serves as a reservoir for long-term carbon storage and the regulation of soil structure, water retention, and nutrient availability, where nitrogen is critical and central to nutrient cycling and plant [73]. The bulk density reflects soil compaction: A higher bulk density, often found in urbanised areas, reduces porosity and limits root growth and microbial [53][54]. Soil pH influences the composition of the microbial community and the enzymatic processes, and it has been proven that urban soils frequently exhibit altered pH due to construction materials and pollution, which in turn affects the nutrient [76].). Electrical conductivity provides a measure of soil salinity or ionic strength, which influences nutrient uptake by plants and soil organisms, particularly in degraded or engineered urban areas [77]. Together, these indicators are critical because they link physical and chemical soil conditions with ecosystem processes such as carbon sequestration, nutrient cycling, and plant productivity, making them indispensable for assessing how urbanisation degrades or transforms ecological ground conditions [78]).

1.1.9 Urban Ecology

City or urban ecology examines how cities, as a novel of ecosystems that host human activities and profoundly shape the ecological structure and processes, take place. It is characterised in large part, among other indicators, by a mosaic of built infrastructure, green patches, remnant fragments, and highly modified

environments, which create complex landscapes with unique microclimates and resource distributions together [79].). Fragmentation in a real sense and according to [55][56] is a pervasive outcome of urbanisation where natural habitats are broken into smaller, isolated patches by roads, buildings, and other infrastructure, species movement, genetic flow, and ecological connectivity decline, contributing to biodiversity loss and functional homogenisation. Anthropogenic pressures in the context of ecosystems go beyond fragmentation, they include the following: pollution (air, light, noise), invasive species, and altered disturbance regimes, all of which reshuffle community composition toward generalist and disturbance-tolerant species [44][57]2016).). In tropical cities, where there are urban greening opportunities, but planning is often weak, we can see that these pressures combine to cause significant changes in ecological dynamics, making urban ecology a critical framework for understanding how human-dominated landscapes affect plant diversity, ecosystem functioning, and long-term environmental resilience.

1.2 Review of Related Concepts and Approaches

1.2.1 Detection of land use / Land Cover changes

LULC change detection is a cornerstone approach in ecological and urban research, involving the use of satellite imagery and remote sensing tools to effectively monitor and evaluate how far human activities and natural processes reshape the landscape over time and place. By leveraging multi-temporal satellite data (Landsat, Sentinel), researchers can map changes in built-up areas, vegetation, water, agriculture, and other classes of LULC across decades, offering a retrospective view of urban expansion and ecological [58][59].). Two of the most widely used change detection techniques/approaches are post-classification comparison, where independently classified images from different dates are compared to produce “from-to” change maps, and supervised classification, in which training data (from ground truth or high-resolution imagery) guide the labelling of pixels in discrete land-use classes [60][61].). The post-classification methods often includes constructing transition matrices, which solely quantify how much area in size and percentage has converted from one class (forest) to another (built-up), allowing for clear interpretation of patterns such as urban sprawl or habitat loss [87]. At a global scene, such methodology have demonstrated effectively how growing urban cities expand in extent, rising impervious surfaces and leading to consequential loss of vegetated and cropland; locally, studies in Nigeria have employed supervised classification to document dramatic built-up growth at the expense of vegetation over multi-decade

periods, underscoring the relevance of change detection for regional ecological assessment and policy-[62][63]).

1.2.2 Assessment of plant diversity in Urban settings

1.2.2.1 Vegetation Structure in Urban Landscapes

The plant and vegetation structure in urban landscapes reflects a complex interplay of species composition, disturbance regimes, and environmental modifications associated with urban development in totality. Species composition in cities typically consists of a mix of native species, ornamental plantings, and non-native or invasive species that thrive under altered conditions, leading to biotic homogenisation and reduced ecological uniqueness across urban grids which is as a result the high human population [64][65].). Urban environments are shaped by disturbance gradients known as (rural, peri-urban, and urban) that range from low-intensity disturbances in urban peri-urban and suburban green spaces to severe disturbances in urban centres where impervious surfaces, pollution, and frequent human activity impose strong filtering pressures on plants [92].). As a result, urban flora (plants) tends to be dominated by disturbance tolerant, fast-growing generalist species with broad ecological amplitudes, while sensitive and specialist species decline in abundance or disappear entirely [66][67].). In tropical cities, including those in the Sub-sahara region, the vegetation structure is purely and additionally influenced by informal land-use practices, high population densities, and weak enforcement of green-space policies, which is proven to have contributed to fragmented vegetation patches, reduced canopy cover, and shifts in community composition along the rural-urban divide [95].).

1.2.2.2 Biodiversity Indices

Plant biodiversity indices are essential tools used to quantify patterns of species richness and distribution in ecological studies, particularly within human-modified environments such as urban gradients (the rural, peri-urban, and urban). The Shannon-Wiener Index (H') measures diversity by incorporating 1) Shannon, 2) Simpson, and 3) evenness indices to track the trend of biodiversity loss, identify dominance species patterns, and compare the whole plant ecosystem structures along rural-urban gradients, demonstrating their reliability and relevance for understanding how urbanisation alters ecological integrity [68][69].).

1.2.2.3 Statistical Analysis of Plant Diversity

Comparing plant diversity across urbanisation gradients is often based on one-way analysis of variance (ANOVA) to test whether mean diversity indices (Shannon H', Simpson's <1-D, evenness) differ significantly between zones (rural, peri-urban, urban), as it partitions variance between-groups and evaluates if between-group variation exceeds within-group noise, as perfectly described in this study [98].). However, the validity of ANOVA depends on key assumptions: 1) normality of residuals, 2) homogeneity of variances, and 3) independence of observations which are frequently violated in ecological community data, especially for diversity indices derived from species counts [99].). To address this, ecologists first perform diagnostic tests such as 1) Shapiro-Wilk test for normality and 2) Levene's test (or Bartlett test) for equal variances; if assumptions hold, standard ANOVA followed by post-hoc comparisons (Tukey's HSD) is appropriate [100].). When these assumptions are not met, non-parametric alternatives such as the 1) Kruskal-Wallis test (a rank-based analogue to 2) one-way ANOVA) provide a robust method for comparing medians or distributions across more than two groups without assuming normality on comparative scales such as urban gradients shown in this study [101].). This approach has been successfully applied in urban ecological studies: for example, researchers comparing spontaneous plant species richness along an urbanization gradient used Kruskal-Wallis to detect significant differences in [102]. By combining 1) parametric (ANOVA) and 2) non-parametric (Kruskal-Wallis) tests in diversity analysis, the research ensures methodological rigor, which accommodates non-normal ecological data, and enhances the reliability of inferences about how urbanization influences plant community diversity.

1.2.3 Ecosystem Functioning Across Urban Gradients

Urban growth significantly alters 1) ecosystem productivity by reshaping thermal environments, 2) soil properties, and 3) vegetation health. Thus, studies along urban-rural gradients show that plant photosynthesis and productivity, often measured via proxies like 1). GPP_{max} and 2). EVI_{max} declines with increasing urban intensity, indicating that dense development suppresses vegetation functionality as proven in this study [103]. For example, in the Yangtze River Delta, urbanization was found to reduce photosynthetic capacity in forests, grasslands, and wetlands, with both direct and indirect effects constraining vegetation growth. Interestingly, the study demonstrated that indirect effects such as reduced surface area can sometimes offset or worsen direct losses in productivity [104].

Soil degradation is another critical consequence. Various research works show that urbanization leads to a dractical reduction in soil organic carbon (SOC) and nitrogen along urban gradients, which also leads to the disruption of nutrient cycling and soil structure as reported in this study [105].). This deterioration undermines the ability of soil to support plant growth, store carbon, and maintain microbial activity. In line with this, it is evidently proven in a more recent work on soil bacterial communities that lower total organic carbon and nitrogen are found in highly urbanised zones, often accompanied by increased soil pH and accumulation of heavy metals, which can further impair ecological processes [106].).

In addition, the effect of the urban heat island (UHI) exacerbates the stress on vegetation in cities. Higher land surface temperatures in urban centres reduce evapotranspiration and increase plant heat stress, while lower vegetation cover and water content amplify thermal discomfort [107]. Possibly, in tropical and humid cities, the cooling effect of vegetation also reverses under certain conditions a dense, physiologically active foliage can increase humidity and heat stress at night, showing complex interactions between plant function and microclimate [108].).

Lastly, some literature suggested that warming induced by urbanisation can unintentionally boost vegetation growth under specific thermal conditions [109]. For example, in rapidly urbanizing agglomerations, rising temperatures have been connected to increased net primary productivity (NPP), driven largely by the thermal environment rather than other environmental factors [109]. This paradox underscores the nuanced role of urban warming and can also degrade ecosystem functioning in many contexts, under certain thermal regimes it might enhance productivity, revealing a complex, context-dependent relationship between urbanisation and ecological processes.

1.2.4 Vegetation Indices in Urban Ecology Research

Vegetation indices such as NDVI and EVI are widely used in urban ecological research to monitor greenness, productivity, and vegetation health in response to changes in land-use. The NDVI is a fundamental biodiversity index derived from red and near-infrared (NIR) reflectance; where high NDVI values typically indicate dense, photosynthetically act NDVIation, while low NDVI values point to sparser or stressed vegetation cover [110].). However, in typical densely vegetated or built-up urban settings, Enhanced Vegetation Index often outperforms NDVI because it includes a blue-band correction to reduce atmospheric and soil background noise,

making it more reliable for measuring vegetation in complex or high-biomass areas as typically highlighted in the study by [111].). Many researchers have used both indices over the years to detect vegetation loss over a long and short period of time, map “green space” dynamics and assess the impact of urban expansion: for example, declines in NDVI have been correlated with increased impervious surfaces in tropical and temperate cities, signalling ecological degradation [70][71].). In addition, some studies have connected long-term trends in NDVI and EVI with land-use transformation, showing that urban greening or degradation can be quantified and spatially mapped to guide city planning and sustainability efforts [72][73].). Collectively, when used together, these indices provide robust and scalable measures for assessing ecosystem health in urban environments, particularly when field measurements are impractical or limited, as studied by Rumora et al. 2021.

1.2.5 Soil Quality and Urbanization

Urban growth in its real form exerts a strong pressure on soil quality through construction activities, land conversion, and the proliferation of impervious surfaces, all of which alter the physical and biochemical properties of soils and diminish their ecological function. Replacement of natural vegetation with buildings, roads and compacted surfaces (human activities) has a significant influence on organic matter input, disrupts soil aggregation, and accelerates erosion, thus lowering soil organic carbon (SOC) and total nitrogen, two core indicators of soil fertility and ecosystem functioning [74][75].). Various and past studies, in both tropical and subtropical cities have revealed that SOC and nitrogen were significantly reduced along urbanisation gradients as vegetation cover decreased and soil disturbance intensified, as well as described in this study [118].). In addition, urban soils frequently exhibit a higher bulk density due to heavy machinery compaction for construction purposes, foot traffic, and infrastructure load; this in turn reduces porosity, infiltration rate, and root penetration, ultimately affecting water retention and nutrient cycling, as well indicated and pointed out by this study [76][77]. The cumulative effects of these processes shape the structure of the plant ecosystem by favouring disturbance-tolerant, shallow-rooted, or invasive species and suppressing native or specialist species that require stable, nutrient-rich and well-aerated soils [78][79]. Consequently, soil degradation becomes a major driver of biodiversity loss and diminished ecosystem resilience in rapidly expanding tropical urban environments around the world.

1.2.6 Urbanization Indicators and Ecological Impacts

Urban growth indicators such as built-up density, night-time lights, and population density, among others, provide powerful quantitative indicators for assessing the intensity of human activities and their ecological consequences in rapidly changing landscapes across the globe [123]. Built-up density or percentage which is derived from land-cover classification or impervious surface mapping captures the spatial extent of construction and infrastructure expansion, which often correlates with habitat loss, vegetation decline, and soil [123]. Night-time lights (NTL), particularly derived from VIIRS-DNB and similar sensors, serve as reliable proxies for 1) anthropogenic intensity, 2) urban energy consumption and 3) socio-economic activity, which have been widely used to model ecological stress, biodiversity loss, and surface temperature dynamics in urban ecosystems as outlined and stressed by [80][81]. Population density, on the other hand, remains one of the most widely used measures of human pressure, as the increase in human population concentrations intensifies land demand, increases pollution, and amplifies disturbances that reduce ecological integrity [126]. Studies in global and some part Asia, some part in South America, and Europe have demonstrated stronger associations between the values of the high urbanisation indicators and reduced vegetation cover, altered ecosystem functioning, and fragmentation of green spaces [82][83]. African case studies including those in Nigeria, Ghana, Kenya, and South Africa consistently show and have also proven beyond doubt that increased built-up density and night-time luminosity correspond to decreased NDVI/EVI values, rising land surface temperatures, declining soil fertility, and reduced species richness, emphasising the vulnerability of tropical and sub-Saharan ecosystems to rapid urban expansion [84][83].

1.2.7 Statistical and Modelling Approaches in Urban Ecology

Statistical modelling plays a central and critical role in urban ecology studies by enabling researchers to quantify relationships between ecological indicators and urbanisation metrics (indicators). One out of many of these statistical models is the Multiple linear regression which remains one of the most widely used tools for examining how environmental variables respond to anthropogenic pressures also known as (human activities), allowing ecological studies to interpret the direction and magnitude of predictor effects through model coefficients while controlling for collinearity and confounding [130]. In most recent vegetation studies, regression models were frequently used to connect NDVI with urban predictors to provide

effective and reliable insights into how urban intensity and expansion have driven the most changes in vegetation productivity and ecosystem functioning [85][86].). Beyond classical statistics, machine learning innovations, particularly the Random Forest (RF) has become a preferred technique for most remote sensing-based land-use/land-cover (LULC) classification due to its non-parametric nature, ability to handle complex nonlinear relationships, and robustness in the presence of noisy, high-dimensional [133].). RF classifiers have routinely outperform traditional well known and used supervised classification approach such as maximum likelihood or minimum-distance algorithms by generating 1) higher accuracy, 2) reducing overfitting through ensemble averaging, and efficiently ranking variable importance during classification as highlighted by the studies commissioned by [87][26].). Lastly, in a most recent study conducted in tropical and sub-Saharan African cities, RF has demonstrated that it provides superior performance for mapping built-up areas, vegetation classes, and peri-urban mosaics, making it especially valuable for analysing urban expansion and its ecological impacts [135].).

1.3 Theoretical Framework

1.3.1 Landscape Ecology Theory

The Landscape Ecology Theory provides a foundational lens for understanding how the spatial patterns within a landscape have influence and impacted key ecological processes, particularly in rapidly the face of rising urbanising environments. The most central and critical part of this theory is the patch-matrix-corridor model, which ideally conceptualised landscapes as mosaics of habitat patches embedded within a broader matrix and connected by ecological corridors, all of which shape species distribution, gene flow, and ecosystem functioning [136].). Urban growth disrupts this configuration by increasing fragmentation, reducing patch size, and the absolute alteration of connectivity, often creating isolated green spaces surrounded by impermeable built-up matrices that restrict plant dispersal and diminish, as documented in this study conducted by [137].). The theory emphasises spatial heterogeneity the variation in landscape elements across space which becomes more pronounced as urban growth introduces sharp ecological gradients between rural, peri-urban, and urban [136]. These observable spatial changes are closely related to the LULC dynamics that took place over time and space, where the expansion of built-up surfaces transforms 1) natural habitats, 2) reduces vegetation cover, and 3) modifies microclimates, ultimately influencing the composition, diversity, and

resilience of plant communities [138].). In this way, the Landscape Ecology Theory provides a robust framework for the interpretation of how spatial restructuring that is driven by urbanisation (urban expansion) shapes ecological patterns observed in Makurdi and Otukpo.

1.3.2 Species Area Relationship (SAR) Theory

The SAR theory, on the other hand, posits that species richness increases with habitat area and, conversely, declines as the habitat size is reduced; this is a principle that is highly relevant to urban ecological studies where land conversion and fragmentation are prevalent and evident. According to Pan 2013, in the urban environments scenario, progressive reduction of natural habitats through construction, infrastructure expansion, and other landscape modifications often leads to measurable declines in species richness, as smaller and isolated patches can support fewer species and are more vulnerable to ecological disturbances [139].). SAR as a reliable theory provides a predictive framework for estimating biodiversity loss along the rural-urban gradient scale, which is achieved by quantifying how decreasing vegetated area and increasing impervious surfaces alter the capacity of landscapes to sustain diverse plant communities [140].). The application of this theory in urban ecology has proven valuable for 1) anticipating threshold effects, 2) identifying patches at risk of local extinction, and 3) guiding green-space planning to maintain minimum habitat sizes necessary for ecological stability [136].). In the context of tropical and African cities, recent studies have confirmed that SAR-based predictions align closely with observed declines in native plants, highlighting the close sensitivity of plant diversity to spatial reduction and also the importance of conserving continuous or connected green areas within expanding urban matrices [141].).

1.3.3 Intermediate Disturbance Hypothesis (IDH)

The third theory (IDH) proposes that species diversity is maximized at moderate levels of disturbance, whereas very low or very high disturbance results in reduced diversity due to competitive exclusion or severe habitat alteration, respectively. This concept is highly relevant to urban ecological studies, where disturbance intensity typically increases from rural to peri-urban to urban zones. Moi et al., 2020, in their study, stated that the Peri-urban areas often exhibit the highest plant diversity because they experience intermediate disturbance levels enough to prevent

dominance by a few competitive species, yet not so intense as to eliminate habitat structure or resource availability. In contrast, Fox 2013 in his study also deduced that rural areas with minimal disturbance may Favor competitive exclusion leading to lower functional redundancy, while heavily built-up urban cores often show simplified plant communities dominated by disturbance-tolerant or invasive species. In their study also, Sheil and Burslem 2013, suggested that applying IDH to urban gradients provides a theoretical explanation for observed patterns of species richness, especially in tropical cities where land-use change creates heterogeneous mosaics of agricultural fields, fallow lands, remnant vegetation patches, and built-up. A recent empirical study on urban landscapes confirms that intermediate disturbance conditions typical of peri-urban zones frequently support the highest levels of herbaceous and woody plant diversity, validating IDH as a useful framework for interpreting biodiversity responses to urbanization [16].).

1.3.4 Urban Scaling Theory

UST posits Urban Scaling Theory (UST) provides a quantitative framework for understanding how different characteristics of cities such as population size, built-up density, infrastructure demand, and resource consumption scale in predictable, nonlinear ways with urban growth (). According to UST, many urban indicators follow power-law relationships, where biological, economic, and infrastructural variables do not increase proportionally but instead exhibit superlinear or sublinear scaling depending on the nature of the process involved [88][89].). For ecological studies, this theory is especially relevant because rapid increases in population density and anthropogenic activity often produce disproportionately large environmental impacts, including intensified land conversion, highly heat island effects, and accelerated resource depletion as cities tend to grow beyond certain limit according to the study by [144].). Urban scaling therefore provides a conceptual lens for interpreting why ecological pressures such as vegetation loss, soil degradation, and biodiversity decline tend to escalate more rapidly in larger cities than in smaller settlements [145] Similarly, in the context of urban gradients, UST helps to explain how increases in population density/growth and built-up density can lead to nonlinear ecological responses, highlighting the importance of quantifying both demographic and spatial indicators when assessing ecosystem quality in urbanizing landscapes with the visible urban gradients.

1.3.5 Ecosystem Function Theory

Ecosystem Function Theory explains how biological communities regulate key ecological processes such as elementary productivity, nutrient cycling, decomposition, soil formation, and energy [90][15]. Central to the theory is the idea that biodiversity, particularly plant diversity, plays a fundamental role in stabilizing and enhancing these ecosystem functions just like Loreau 2000 highlighted in his study. According to Daily and Matson 2008, the theoretical foundation is vital for understanding how vegetation patterns shift along visible urbanization gradients, where biodiversity loss often translates into reduced ecosystem functioning, poorer soil conditions, and diminished ecological resilience.

The theory also provides the conceptual basis for linking remote sensing indices such as NDVI and EVI to ecological processes. Since NDVI and EVI capture differences in chlorophyll abundance, canopy density, and vegetation Vigor, they serve as proxies for photosynthetic activity and aboveground productivity, key aspects of ecosystem functioning. Daily and Matson 2008 state that, in areas with higher plant diversity, productivity tends to be more stable and robust, which is often detectable as consistently higher (NDVI/EVI). In contrast, in highly urbanized areas where vegetation is sparse or stressed, these indices typically decline, reflecting an impaired ecosystem [147]. By integrating biodiversity assessments with spectral indicators, researchers can now assess how these urban growth and expansion have modified ecological processes across spatial gradients, making Ecosystem Function Theory a critical foundation for studies linking plant diversity, soil conditions, and remote-sensing-based vegetation indicators [147].).

1.4 Empirical Review

1.4.1 Global Studies

LULC change driven by urban rising population density is a major global force reshaping habitats and driving biodiversity loss on local, regional and global scale; multi-temporal remote sensing studies consistently show that expansion of built-up and impervious surfaces replaces vegetation and fragments formerly continuous habitats, with measurable declines in native species richness and habitat heterogeneity[91][92],<https://ssrn.com/abstract=4579821>. In studies by [93][94] it was observed that Large-scale analyses using decades of satellite data showed that

urban growth alters landscape composition both directly (conversion of natural land to built surfaces) and indirectly (changes in hydrology, microclimate and disturbance regimes), producing “from-to” transitions—forest → agriculture → built-up or cropland → built-up that can be quantified with post-classification change matrices to reveal the magnitude and pathways of habitat. Previous studies by [95][96] observed and stated that global syntheses further indicate that the ecological consequences of these transitions are not uniform: tropical regions often experience especially rapid LULC change and steeper biodiversity declines because baseline species pools tend to be richer and more sensitive to habitat area reduction and fragmentation.

The effects of urbanization on vegetation productivity are complex but increasingly well documented through vegetation indices (NDVI, EVI) and gross primary productivity proxy. Large-sample remote sensing studies spanning hundreds of cities, to which these belong [153], have found systematic reductions in vegetation greenness and seasonal productivity within urban cores relative to surrounding rural areas, although responses can be regionally variable and influenced by climate, greening efforts, and temporal trends in land management. In particular, recent works by [97][98] after separating the direct effects (loss of vegetated area) from indirect effects (altered microclimate, edge effects) discovered and pointed out that both pathways contribute to observed drops in productivity; in many cities the net effect is a measurable decline in NDVI/EVI associated with increasing impervious cover and anthropogenic disturbance. At the same time, study by [153] strictly cautions that urban vegetation dynamics are heterogeneous where active greening or irrigation regimes exist, NDVI trends may locally increase despite overall habitat loss, so interpretation requires integrating spectral data with ground-truth biodiversity and land-use information.

Soil degradation under urban pressure is another global pattern with clear implications for ecosystem functioning. Empirical studies and meta analyses from the following researchers [99][100] have documented declines in soil organic carbon (SOC) and total nitrogen, increases in bulk density (compaction) and altered pH and salinity in urban cores compared to changes less disturbed in peri-urban and rural soils, which reduce soil fertility, water infiltration, and microbial activity. Taken together, as opined by Çelikgil et al. 2025, in their study therefore emphasize that LULC change, declines in vegetation productivity (NDVI/EVI), and progressive soil degradation form an interlinked suite of processes through which urbanization undermines biodiversity and ecosystem functioning highlighting the need for

integrated, multi-scale analyses that combine remote sensing, field ecology, and soil science.

1.4.2 African Studies

In their work [156], suggested that African cities are increasingly the focus of empirical urban ecology research, revealing the recurring patterns in biodiversity, remote-sensing applications, vegetation-index trends, and soil change driven by rapid urban expansion. Also [156] study have synthesized urban biodiversity patterns across the continent and reported consistent outcomes: 1) that the urban cores and highly built-up neighborhood's tend to be dominated by disturbance-tolerant and 2) that the non-native species, the peri-urban mosaics often retain higher species richness which support the intermediate-disturbance expectation, and 3) that the fragmentation and the loss of contiguous natural habitat drive declines in specialist taxa and functional diversity. In the systematic review [157], it was highlighted that strong geographic biases in African urban ecology (many studies concentrated in southern and western African capitals) while confirming common patterns of homogenization and reduced native species representation in city center's, which was also supported by [156].). According to [157], Field studies in secondary cities in West Africa similarly document shifts toward generalist flora and loss of native understory species as built cover increases.

Remote sensing has become indispensable for scaling and filling data gaps in African urban studies. Multi-temporal Landsat and Sentinel time series, often classified with Random Forest or other ensemble methods, have been used to map built-up expansion, quantify impervious-surface growth, and generate transition matrices that reveal pathways of vegetation → urban conversion across West and Sub-Saharan Africa which have been heavily documented in these studies by the researchers [101][102] and by extension they have also in their works produced standardized multi-temporal built-up maps for dozens of Sub-Saharan cities, by demonstrating how fused satellite products can robustly document two decades of urban growth even where ground data are sparse. Weldegebriel et al., 2021, in their work, opined that these remote-sensing products are frequently combined with population rasters (WorldPop) and night-time lights to derive multi-indicator urbanization layers that inform ecological analyses and planning in rapidly changing African contexts.

Time-series NDVI/EVI analyses in African cities provide spatially explicit evidence of vegetation loss, recovery, and heterogeneity. A regional based study conducted by [160] shows that systematic NDVI declines within dense urban cores (expanded cities) compared with the surrounding rural areas, although localized greening initiatives (parks, managed gardens, irrigated crops) can produce positive NDVI anomalies that complicate simple urban vs rural contrasts in the study region or anywhere else. a couple of research by these authors' [103][104], opined for example, that NDVI analysis paired with field validation in several West African cities identified peri-urban “refugia” where vegetation remains relatively intact and could be prioritized for conservation or green-infrastructure intervention. Such combined remote-sensing and field approaches strengthen inferences about the ecological consequences of urban expansion and help target areas for restoration.

Similarly, these researchers in their work [105][106], carried out soil studies across African urban gradients, although fewer in number than remote sensing papers consistently document degradation trends associated with urbanization: declines in soil organic carbon (SOC) and total nitrogen, increased bulk density indicative of compaction, and altered pH/EC profiles that can impede plant growth and microbial. In addition, below-ground changes are frequently linked to above-ground vegetation loss and imperviousness as observed by [163] in their work; for example, a study in Central and Southern Africa reports lower SOC and poorer physical soil quality in highly urbanized zones versus peri-urban and rural sites, with management history (irrigation, organic amendments) modulating local patterns [163].).

1.4.3 Nigerian Studies

In the context of Nigeria, the urban landscape has experienced a sustained and rapid increase in human population and its related characteristics over recent decades, with major metropolitan areas and numerous secondary cities undergoing a lot of spatial growth that often outpaces planning capacity. Davis et al. 2023 in their most recent work documented from the findings of their work a pervasive conversion of agricultural and vegetated land to built-up uses, increasing impervious surfaces, and fragmenting natural habitat trends that have important implications for biodiversity and ecosystem services in the Nigerian landscape in the year 2023. Also, previous literatures in the National and regional assessments (Oladunmoye 2024; Barau et al. 2015), illustrate that the urban growth patterns in Nigeria are mostly heterogeneous

(the coastal megacities vs. the inland secondary cities), but the common outcome is accelerated landscape transformation that reduces contiguous green areas and intensifies land-use competition, particularly around fast-growing administrative and transport hubs.

Empirical studies in Nigeria reveal consistent signals of loss of biodiversity and community homogenization in urban centers. Field and plot based surveys that was recently conducted by [10], indicate in their findings that the urban centers frequently host lower native species richness and higher proportions of disturbance-tolerant, opportunistic, or exotic taxa compared with peri-urban and rural sites; peri-urban mosaics often retain the highest local richness consistent with intermediate disturbance dynamics. Several Nigerian case studies, viz [9][109][110], have combined plot inventories with LULC mapping to discover the declines in understory and specialist flora as built cover increases, underlining the urgent need for targeted conservation of remnant patches and peri-urban refugia seen in the Nigerian landscape.

Soil studies and ecosystem assessments from Nigerian contexts corroborate the patterns of below-ground degradation associated with urbanization. A research conducted across urban gradients and in converted peri-urban lands have a report clearly stating the reductions in soil organic carbon and alterations in nitrogen pools, together with increased bulk density and disrupted infiltration changes that undermine soil fertility and the ability of urban soils to support diverse plant communities and sustain ecosystem services such as water regulation and carbon storage [168], <https://emergingsociety.org/index.php/efltajssei/article/view/413>. NDVI/EVI-based investigations in Nigeria have been increasingly used to track vegetation trends and to identify hotspots of greenness loss or localized greening initiatives. Lemenkova and Debeir, 2023, stated that the declining greenness in the urban cores that are related to peri-urban and rural areas, while targeted urban greening projects (parks, institutional gardens, riparian buffers) produce detectable positive anomalies that are useful for conservation planning. Integrating spectral indices with field plots and LULC change matrices has proven especially informative for secondary cities (the kinds of places studied here), allowing identification of priority areas where green-space protection or restoration would most effectively conserve biodiversity and improve ecosystem functioning [169].).

1.5 Summary of Research Gaps

Despite increasing interest in urban ecological research, several gaps remain evident in the Nigerian context, particularly within the Benue State. First, on the list, only a limited number of studies connected remote sensing datasets with field-measured plant diversity indicators and soil ecosystem properties, leaving an incomplete understanding of how urbanization simultaneously alters vegetation structure (the whole plant ecosystem) and soil function. The Second on the list is that there is a notable scarcity of studies applying NDVI and EVI regression models combined with urbanization metrics such as built-up density, population density, or night-time lights, although these tools are essential for quantifying ecological responses along urban gradients. Third on the list is the fact that fine-scale vegetation plot data are largely unavailable for Benue State, and past studies in Makurdi, Otukpo, and surrounding areas have mostly relied on coarse land-cover maps or rapid assessments, offering little insight into species richness, evenness, or community composition. Fourth, on this list, there is also limited research that evaluates soil organic carbon, nitrogen, bulk density, pH, and EC together as ecosystem indicators across urban-rural transitions, resulting in weak empirical connexion's between soil quality and vegetation patterns. And finally on this list, it is a very observable fact that very few studies in Nigeria explicitly test the biodiversity-ecosystem functioning relationship, leaving a major gap in understanding how plant diversity translates into productivity, vegetation health, and overall ecological resilience under rapid urban expansion. The current study tackled and addressed these gaps by combining remote sensing, field biodiversity surveys, soil analyses, and statistical modelling to generate comprehensive evidence for urban ecological change in Benue State.

1.6 Conceptual Framework

The conceptual framework behind this study, as seen in Figure 1, illustrates how urbanization initiates a chain of ecological transformations that ultimately influence ecosystem functioning across urban–peri-urban–rural gradients.

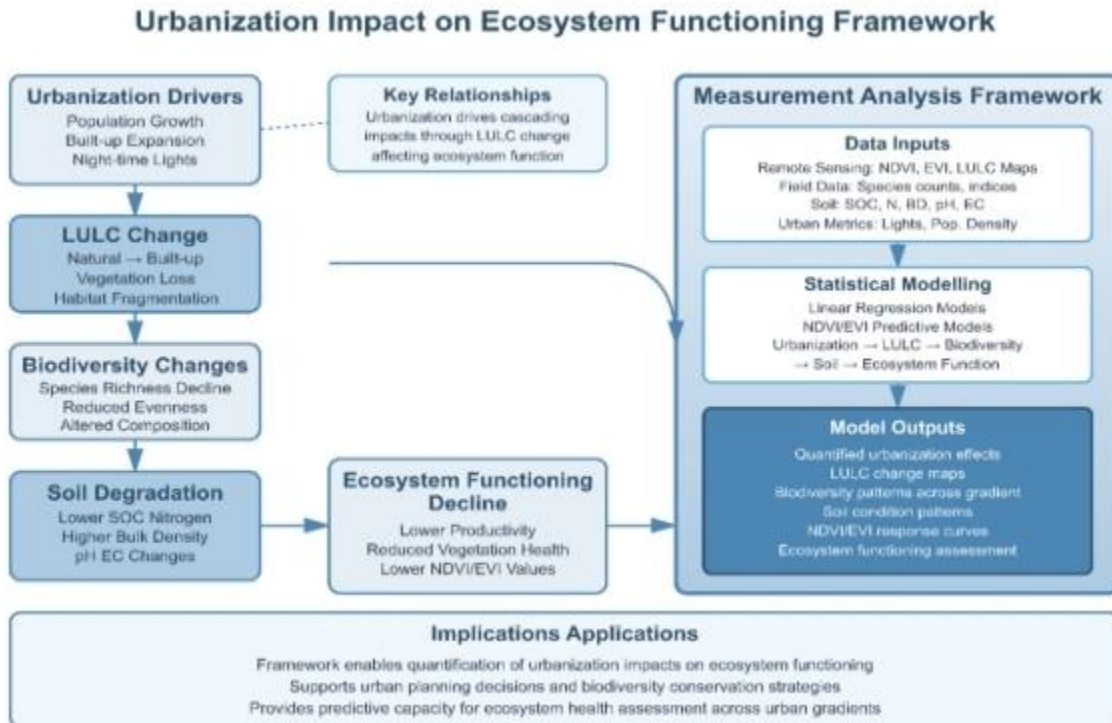


Figure 1.1: Conceptual framework flow chart: source author

CHAPTER TWO

2.0 RESEARCH METHODOLOGY

2.1 Research Design

This study used a multi-scale quantitative ecological research design that combines geospatial analysis with field-based assessments of biodiversity, soil characteristics, and statistical modelling to investigate the influence of urbanisation on ecosystem functioning in Makurdi and Otukpo, as illustrated in Figure 3.1. The research design is grounded in a systematic integration of remote sensing techniques, field-based ecological observations, and inferential statistical analyses, aligning with current approaches in urban ecological research [111][112][113].

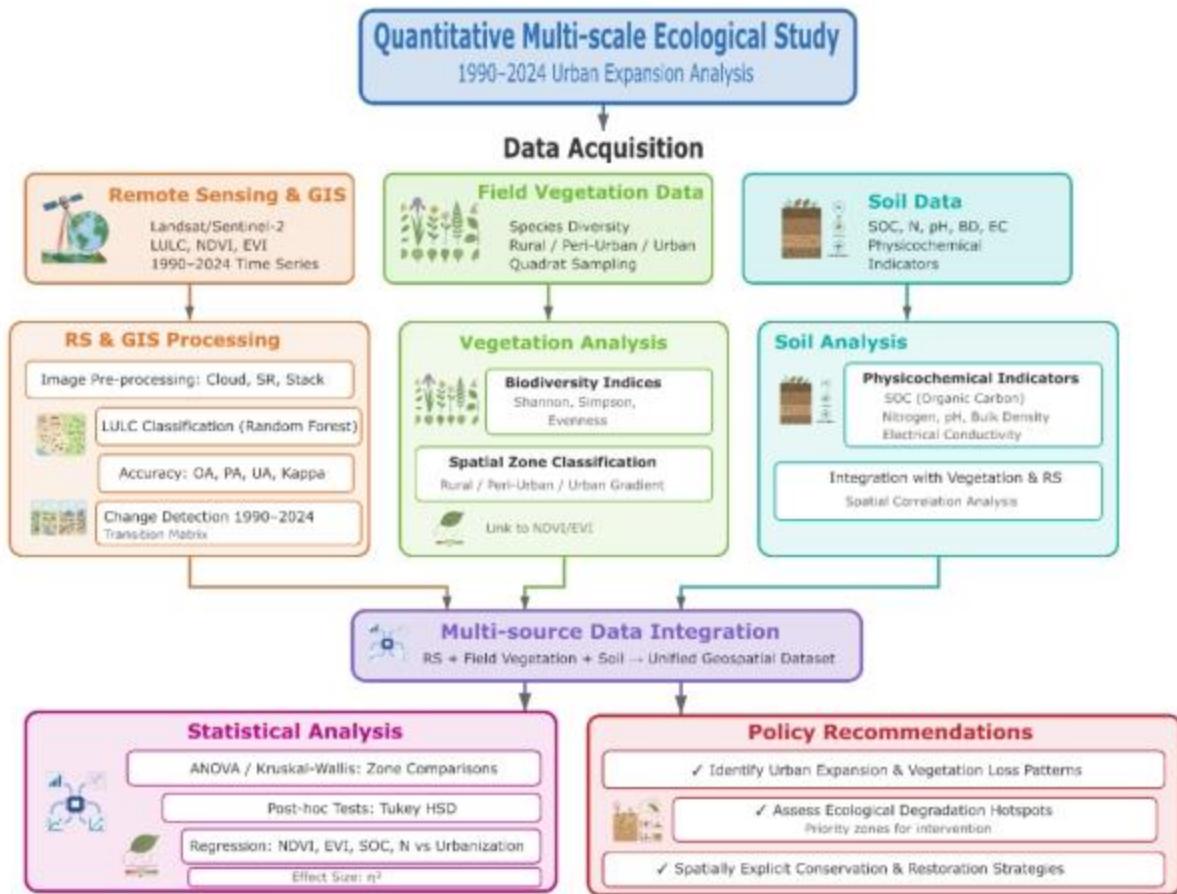


Figure 2.1: Methodological framework flow chart: source of the author

2.2 Study Area

Benue State, in the north central of Nigeria, had a land area of 2,882 km² and a population of 4,253,641. The state was divided into 23 (23) local government areas. Geographically, Benue State lies within the lower River Benue, with longitude coordinates ranging from 7° 47'to 10° 0' East and latitude coordinates from 6° 25'to 8° 8' North (Figure 2.1). The state shares boundaries with five other states: Nasarawa to the north, Taraba to the east, Cross River to the south, Enugu to the southwest, and Kogi to the west. Additionally, the state shares a common boundary with the Republic of Cameroon to the south-east [173]. Benue is characterized by a mix of urban and rural environments, with diverse ecosystems ranging from tropical forests to wetlands and agricultural areas. This study focuses on two cities, Makurdi and

Otukpo. Makurdi Local Government Area is situated in the northeastern part of the State and within the coordinates 7° 00' N to 7° 45' N latitude and 8° 00' E to 8° 32' E longitude. The area is drained by several channels, including the Benue River, which bisects the town into north and south banks, and its tributaries, such as Urudu, Demepe, Kereke, Mu, Idye, and Kpege. According to the 1991 and 2006 population censuses, the population of Makurdi was 239,889 and 297,398, respectively. Using a growth rate of 0.03%, the projected population of Makurdi in 2017 was 517,342. The area has a rapidly growing population of humans, with a density of 323 persons per square kilometre. The population is predominantly Tiv-speaking, with other ethnic groups also present. The growth of Makurdi town has put pressure on natural resources within its environs, including wetlands. Makurdi's climate is characterized by wet and dry seasons. The mean duration of the rainy season is 182 days, with the highest monthly rainfall total of 221 mm recorded in August [174] (see Figure 2.1).

Otukpo is a local government in the Southern Guinea Savannah Zone in central North Nigeria. Otukpo LGA lies on the North of the equator between (the longitude 7° 50' and the 8° 15' East and the latitude 7° 00' and that of 7° 40'). It has a landmass of about 1,385 square kilometres and a population of 266,411 humans according to [175] and was projected to be 359,600 people in 2016. The map of Otukpo LGA is shown in Figure 2.2. The general relief of the study area ranges between 83 m above sea level and 162 m above mean sea level. The study area is a valley given that the area is undulating to gently sloping in the north-south direction. [176] (see Figure 2.2).

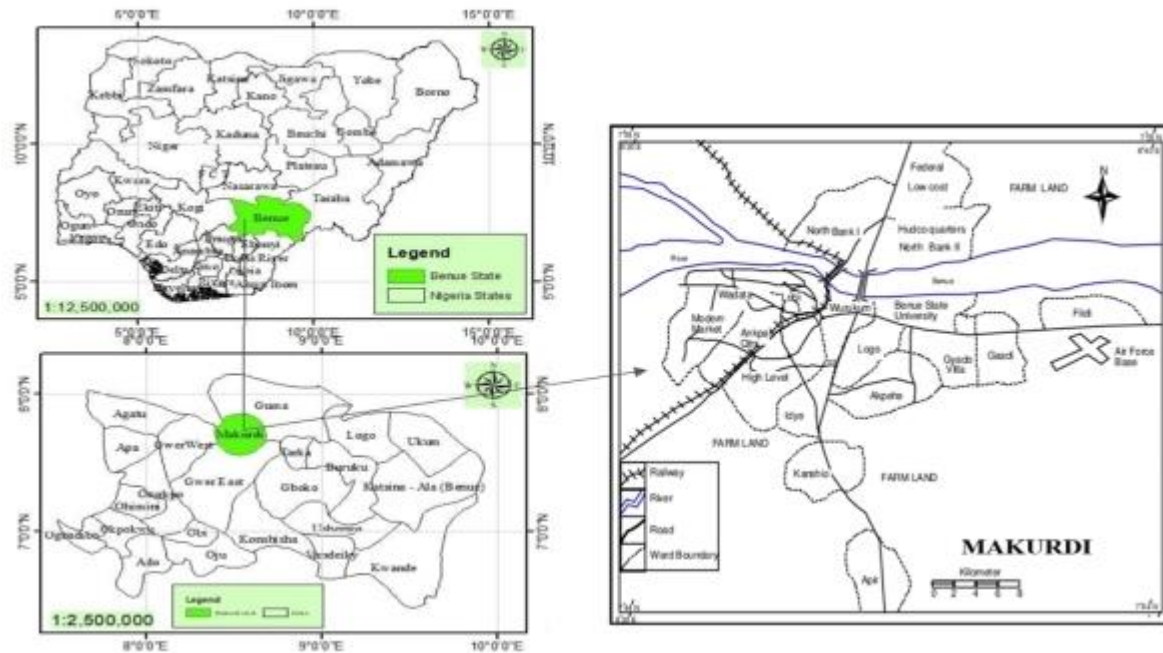


Figure 2.2: Map of the Makurdi Local Government Area

Source [177] & Benue State Ministry for Lands and Surveys.

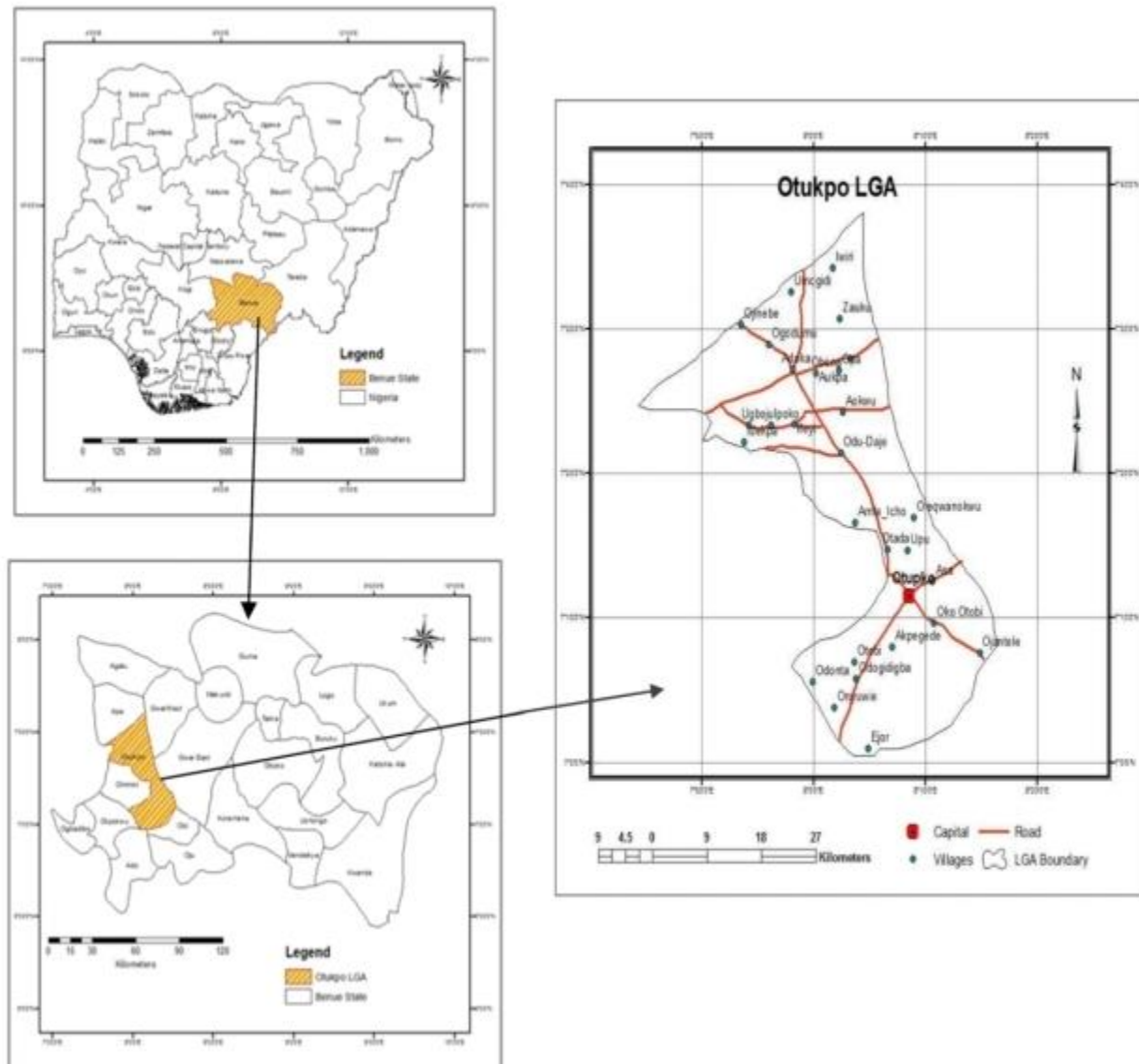


Figure 2.3: Map of the Otukpo Local Government source:[176]

2.3 Data Requirements and Sources

2.3.1 Remote Sensing Data

The study used temporal satellite imagery and ancillary geospatial datasets necessary to detect LULC transformations and transitions over time and space, quantify vegetation condition, and assess urbanization intensity at the two study locations Makurdi and Otukpo. Landsat TM for the (1990) period, ETM+ for the (2010) period, and the OLI (2024) scenes were obtained for change detection, NDVI, and EVI derivation due to their 30 m spatial resolution and long-term continuity, which

makes them suitable for landscape-scale ecological studies. VIIRS Day/Night Band night-time lights imagery was used to represent anthropogenic intensity and built-up expansion through illumination patterns, while WorldPop population density rasters provided high-resolution demographic information to quantify population-driven urbanization pressures these were extracted at the ward level using Appendix 1. In addition, the administrative boundary datasets for the wards and LGAs were incorporated to allow spatial disaggregation of the results and zonal analyses. All layers derived from remote sensing were clipped, standardised, and extracted at both ward level (point sampling) and zonal level (rural, peri-urban, urban) to match the scales of ecological assessment.

2.3.2 Field Plant Data

Plant data were obtained through systematic field sampling conducted in the 33 stratified plot samples, resulting in a detailed dataset comprising species inventories, abundance records, and structural attributes for each sampling location within both study regions. To adequately represent the vertical complexity characteristic of tropical ecosystems, the layers—trees, shrubs, and herbs—were documented separately according to the conditions of the site. For each plot, my team of experts observed and documented the geographic coordinates, its elevation, the surrounding land-use characteristics, and other relevant metadata suitable for the study were recorded to ensure precise spatial referencing and seamless integration with remotely detected datasets. These collected and documented field observations for both locations formed the basis for the calculation of biodiversity indices, including Shannon, Simpson, and Evenness, and facilitated comparative analysis of species composition along the rural–peri-urban–urban gradient.



FIG. 2.4: Ground Truth Data Collected During Field Visits on LULC:

Author source

2.3.3 Soil Data

Soil data was obtained using the GEE cloud computing platform to assess the functioning of the ecosystem through the following key biophysical properties; 1) Organic carbon (SOC), 2) total nitrogen, 3) bulk density, 4) soil pH, and 5) cation exchange capacity (CEC), which together provide insight into soil fertility, compaction levels, and overall chemical balance. These soil variables were extracted from globally validated soil datasets suitable for sub-Saharan Africa and spatially matched to vegetation sampling plots. The soil information was then integrated with field-based vegetation data and remotely sensed variables to evaluate how urbanization influences soil quality and related ecological processes along the rural–

peri-urban–urban gradient. The GEE workflow used for extracting the soil variables is provided in Appendix 1.

2.4 Land-Use and Land-Cover (LULC) Analysis

2.4.1 Image Pre-Processing

All satellite images were pre-processed to ensure radiometric consistency and spatial comparability across the three temporal periods (1990, 2010, 2024). Landsat Surface Reflectance (SR) products generated using LEDAPS for TM/ETM+ and LaSRC for OLI were used, as these datasets provide atmospherically corrected spectral bands suitable for quantitative analysis. Cloud and cloud-shadow contamination were removed using pixel-level quality assurance (QA) masks embedded in the SR products. After the cloud removal was performed on the imageries, the individual spectral bands were combined to form multi-band images for each study year. These images were then clipped to the administrative boundaries of the Makurdi and Otukpo Local Government Areas using official shapefiles, ensuring that the analysis was restricted to the study areas only. And to ensure consistency across time and datasets, all images were resampled to a uniform 30m spatial resolution; this is to allow a more reliable comparison between years and seamless integration with NDVI, EVI, and land-use classification outputs.

2.4.2 LULC Classification Using Random Forest

LULC mapping was performed using a machine-learning algorithm supervised classification approach (Random Forest). This method was selected because of its reliability, strong performance in handling complex, non-linear relationships, and suitability for heterogeneous tropical environments. Training samples for classification purpose were compiled from field-collected GPS points, high-resolution Google Earth imagery, and expert visual interpretation of different features of the land-cover in the study areas.

The Random Forest model was implemented using 500 decision trees, with tuning parameters adjusted to achieve a balance between model accuracy and stability. Five major land-cover classes were defined for the purpose of classification: 1) vegetation, 2) built-up areas, 3) agricultural land, 4) bare surfaces, and 5) water bodies. Following classification, a simple post-classification filtering procedure was applied to reduce the salt-and-pepper effect commonly associated with pixel-based classifications. The final land-use and land-cover maps produced for each study year

were then visually examined to confirm spatial consistency and ensure that the classification results reasonably reflected the conditions of the ground.

2.4.3 Accuracy Assessment

The precision of each LULC map was evaluated through the independent set of validation points versus the classified outputs that were compared against the reference data (see Appendix 2). Each map was evaluated to determine the classification performance using the confusion metric. And the overall accuracy, which reflects the proportion of correctly classified pixels; Producer's accuracy, which measured how effectively real-world LULC types were captured by classification; User's accuracy, which indicated the reliability of each mapped class; and Kappa coefficient, which accounts for agreement beyond random chance were performed in all temporal classifications done.

In total, the accuracy measures provided a robust and effective basis for the evaluation of classifier performance and confirmed the suitability of the Random Forest approach for land-use and land-cover mapping in complex tropical landscapes.

2.4.4 LULC Change Detection

The LULC change was analyzed by comparing the classified maps for 1990, 2010, and 2024. A transition matrix was used to track how land-cover classes changed over the 34-year period, allowing gains, losses, and areas of persistence to be clearly quantified. From the change detection analysis, also known as post classification, key indicators were derived, including the rate of urban expansion reflected by increases in built-up areas, the extent of vegetation loss indicating reductions in natural and semi-natural cover, and shifts in the distribution of agricultural land and bare surfaces.

2.5 Plant Diversity Assessment

2.5.1 Sampling Design

A stratified field sampling approach was used and this is to effectively capture plant diversity across the urbanization zone (rural–peri-urban–urban gradient) in the Makurdi and Otukpo LGAs. Stratification was based on administrative wards, by enabling a fine-scale ecological assessment that reflects actual settlement patterns

and varying levels of urbanization. In the Makurdi LGA, six wards were sampled out of the 11 wards, wherein each ward three plots were sampled, which resulted in a total of 18 plots. Similarly, in Otukpo, five wards were sampled with three plots each, giving 15 plots. In total, 33 field plots (see Appendix 3, 3P1–3P33) were established in the two LGAs to ensure comprehensive and representative vegetation data.

A nested hierarchical plot design was used to capture plant diversity across different layers of vegetation. For trees, a 10 m × 10 m quadrat was set in each plot. Within each tree plot, a subplot of 5 m × 5 m was designated for shrubs, and a subplot of 1 m × 1 m was placed centrally for herbaceous species for consistency purpose. The chosen layout allowed for simultaneous measurement of species composition, density, and abundance across vertical strata. All plant species within the plots were identified in the field using standard botanical guides and voucher specimens were collected for species that required further verification. The abundance of species was recorded for each stratum, and GPS coordinates were logged to support spatial referencing and integration with remote sensing and regression analyses. This sampling framework produced a detailed vegetation dataset suitable for calculating species richness, Shannon-Wiener diversity, Simpson diversity, and species evenness across the urbanization gradient.



Mango tree (*Mangifera indica*)



Coffee senna (*Senna occidentalis*)

FIG. 2.5: Field Work on Plants Species Diversity Data Collection: Author source

2.5.2 Computation of biodiversity indicators

Plant diversity was quantified for each of the 33 plots using three widely applied ecological diversity metrics. These indices provide complementary information on species richness, abundance distribution, and community structure, allowing robust comparison across the urbanization gradient.

Shannon-Wiener Diversity Index (H')

$$H' = - \sum p_i \ln(p_i)$$

Where:

- $p_i = \frac{n_i}{N}$
- n_i = number of individuals of species i
- N = total number of individuals in the plot

This index captures both species richness and evenness, with higher values indicating greater diversity and ecological complexity.

Simpson's Diversity Index ($1 - D$)

$$1 - D = 1 - \sum p_i^2$$

This index emphasises the dominance of species and the balance of the community. Values closer to 1 signify more diverse and less dominated plant communities.

Pielou's Evenness (E)

$$E = \frac{H'}{\ln S}$$

Where:

- S = total number of species recorded in the plot

Evenness values close to 1 indicate an equitable distribution of individuals between species, reflecting stable and undisturbed vegetation communities.

The calculation of these indices provided quantitative measures for evaluating how plant diversity structure responds to increasing levels of urbanization [114][18][29].

2.5.3 Statistical Analysis of Diversity

2.5.3.1 One-Way ANOVA

A one-way Analysis of Variance (ANOVA) was applied to test whether the plant diversity metrics differed significantly between the three urbanization zones urban, peri-urban and rural [179]. The dependent variables included:

- Shannon-Wiener Diversity Index (H')
- Simpson's Diversity Index ($1 - D$)
- Pielou's Evenness (E)

Before the analysis, the ANOVA assumptions (normality and homogeneity of variances) were verified using Shapiro-Wilk and Levene's tests. Where the assumptions were met, ANOVA was used to determine whether the mean diversity values differed between the zones. In the case where the assumptions are violated, the analysis was supported by visual inspection and potential reporting of ANOVA results as supplementary information. A Post-hoc comparisons (Tukey HSD) were conducted where ANOVA yielded significant differences, and this helped in identify specific zone-level contrasts. This statistical approach allowed a rigorous assessment of how urbanization intensity influences species diversity and community structure in Makurdi and Otukpo.

2.5.3.2 Post-hoc Tests

Where the one-way ANOVA indicated significant differences among urban, peri-urban and rural zones, the Tukey's honestly significant difference (HSD) test was used for pairwise comparisons. Tukey HSD identifies which specific pairs of zones differ significantly in terms of:

- Shannon-Wiener Diversity Index (H')
- Simpson's Diversity Index ($1 - D$)
- Pielou's Evenness (E)

This post-hoc procedure controls for Type I error across multiple comparisons and provides robust inference on the specific effects of urbanization intensity on plant diversity patterns.

2.6 Ecosystem Functioning and Urbanization Impact Modelling

2.6.1 Extraction of urbanization metrics

The intensity of urbanization for each sampled ward was quantified using the 500 m buffer polygons created around the centroids of the field-validated ward (Appendix 4). These BUFFER were stored as a single asset (WARD_ZONES_BUFFER) and served as spatial units for zonal statistics extraction in GEE using Appendix 1. Three primary urbanization metrics were computed for each buffer:

1. Build-up Percentage (% Built-up)

Derived from the ESA WorldCover 2021 land-use map. The pixels classified as “Built-up (50)” were extracted and expressed as the mean proportion of built-up area within each buffer.

2. Night-time Light Intensity (NTL)

Obtained from the VIIRS 2024 monthly composites. The mean radiance within each buffer served as a proxy for anthropogenic activity and infrastructure intensity.

3. Population Density

Extracted from the WorldPop 2020 gridded population data set. The mean population density within each buffer was used to represent human pressure.

All extracted metrics were exported from GEE into a unified CSV file and standardized using z-scores before regression modelling to ensure comparability and reduce scale effects.

2.6.2 Ecosystem Function Indicators

Ecosystem functioning indicators were derived for each ward buffer using harmonized remote-sensing and soil datasets (Appendix 1). Indicators were computed as buffer-level means and exported as attributes for each ward.

Vegetation Function Indicators

- **NDVI (Normalized Difference vegetable Index)**

Derived from the surface reflectance Landsat 8 (2024) composite using:

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

- **EVI (Enhanced Vegetation Index)**

Computed using:

$$EVI = 2.5 \times \frac{NIR - RED}{NIR + 6 \cdot RED - 7.5 \cdot BLUE + 1}$$

Soil Ecosystem Indicators (from SoilGrids)

- Soil Organic Carbon (SOC)
- Total Nitrogen (N)
- Bulk Density (BD)
- Cation Exchange Capacity (CEC)
- Soil pH

These variables reflect key proxies for ecosystem functioning, soil fertility, physical properties, and chemical status. All indicators were extracted using GEE's reduce Region's function and exported as a single CSV for analysis (Appendix 5).

2.6.3 Zone Comparison Statistics

To evaluate differences in ecosystem functioning across the urban–peri-urban–rural gradient, each ward was categorized into one of the three zones based on pre-field and Google Earth Pro classification (Appendix 4). The following analyses were performed:

1. One-way ANOVA

Used to test significant differences among the three zones for:

- NDVI
- EVI
- SOC
- Nitrogen
- Bulk Density
- pH
- CEC

2. Post-hoc Tests

Tukey HSD was applied where ANOVA indicated significant differences.

3. Effect Size (η^2)

Eta-square values were calculated to quantify the magnitude of the effects of the zone on ecosystem indicators.

4. Assumptions Cheques

- Normality (Shapiro-Wilk and Q-Q plots)
- Homogeneity of variance (Levene's test)

This step helps to establish whether ecological functioning decreases systematically with increasing urbanization.

2.6.4 Regression Modelling

Multiple linear regression was used to quantify how urban expansion influences plant functioning in the sampled wards. The analysis relied on the integrated ward level data set containing built-up density, night-time light intensity, population density, and the vegetation response variable. Also, to maintain statistical clarity and avoid model inflation, the regression was restricted to a single vegetation functioning index in the NDVI. The decision was guided by the well-established reliability of NDVI as a robust proxy for vegetation productivity, canopy greenness, and photosynthetic condition in heterogeneous tropical landscapes. According to these literatures [115][116], NDVI is less susceptible to saturation and background noise

than any other greenness index when applied across mixed urban to rural mosaics, making it particularly suitable for any urban ecological evaluation.

NDVI was therefore used as the dependent variable in a single ecological response model, while built-up density, nightlights, and population density served as explanatory variables representing the urbanization gradient. The regression structure is expressed as follows:

$$NDVI = \beta_0 + \beta_1(Built - up) + \beta_2(Lights) + \beta_3(Population) + \varepsilon$$

By focusing the analysis on a single theoretically validated indicator, the regression results offer a clearer ecological narrative and a more statistically stable assessment of the impacts of urbanization. To achieve all of these, all statistical analysis were performed in Google Colab Python environment using the following python scripts (Appendixes 6, 7, and 8).

2.7 Policy and Management Recommendations

The Policies and management recommendations for Makurdi and Otukpo were developed by absolute synthesization of the main analytical outputs of the study, including LULC change trends, landscape diversity metrics, NDVI variability, soil condition indicators and regression model relationships between urbanization intensity and ecological variables. The spatial and temporal patterns derived from the LULC analysis provided the researcher with the basis to identify where urban expansion, vegetation loss, or agricultural conversion are most significant. These were complemented by diversity metrics, which revealed the extent of landscape fragmentation or homogenization in rural, peri-urban, and urban zones. The extraction of both soil and NDVI further highlighted areas experiencing vegetation stress, declining soil quality, or ecological degradation, while the regression models quantified the influence of urbanization on vegetation and soil conditions.

These multiple lines of evidence were systematically integrated to formulate targeted, data-driven policy recommendations. Each of these recommendations was anchored directly to specific findings such as areas showing rapid built-up

expansion, high fragmentation, low vegetation Vigor, elevated bulk density, or strong regression-based degradation signals. This adopted approach ensured that the final recommendations are not generic, but spatially explicit, evidence-based, and tailored to the distinct characteristics of rural, peri-urban, and urban environments within both LGAs.

CHAPTER THREE

RESULTS AND DISCUSSION

3.1 Land Use and Land Cover (LULC) Classification Results

3.1.1 Land Use and Land Cover Dynamics

The assessment of the distribution of LULC in the study area for 1990, 2010 and 2024 reveals substantial transformations driven by rapid urbanisation, shifting land demands, and environmental change. Table 3.1 summarises the extent of each land cover class and their corresponding changes across the two intervals (1990-2010 and 2010-2024), as well as the overall 34-year period, while Figure 3.1 shows the graphical trend.

Table 3.1: Summary of the Change in the LULC Area in Makurdi (1990-2024)

LULC Class	1990 (Km ²)	2010 (Km ²)	2024 (Km ²)	Change 1990–2010 (Km ²)	Change 2010–2024 (Km ²)	Total Change 1990–2024 (Km ²)
Build-up	23.17	83.26	143.11	+60.09	+59.85	+119.94
Forest	353.77	281.22	228.84	−72.55	−52.38	−124.93
Cropland	332.24	407.35	172.53	+75.11	−234.82	−159.71
Water Body	19.55	12.54	14.69	−7.01	+2.15	−4.86
Wetland	78.82	23.18	185.82	−55.64	+162.64	+107.00

Author source

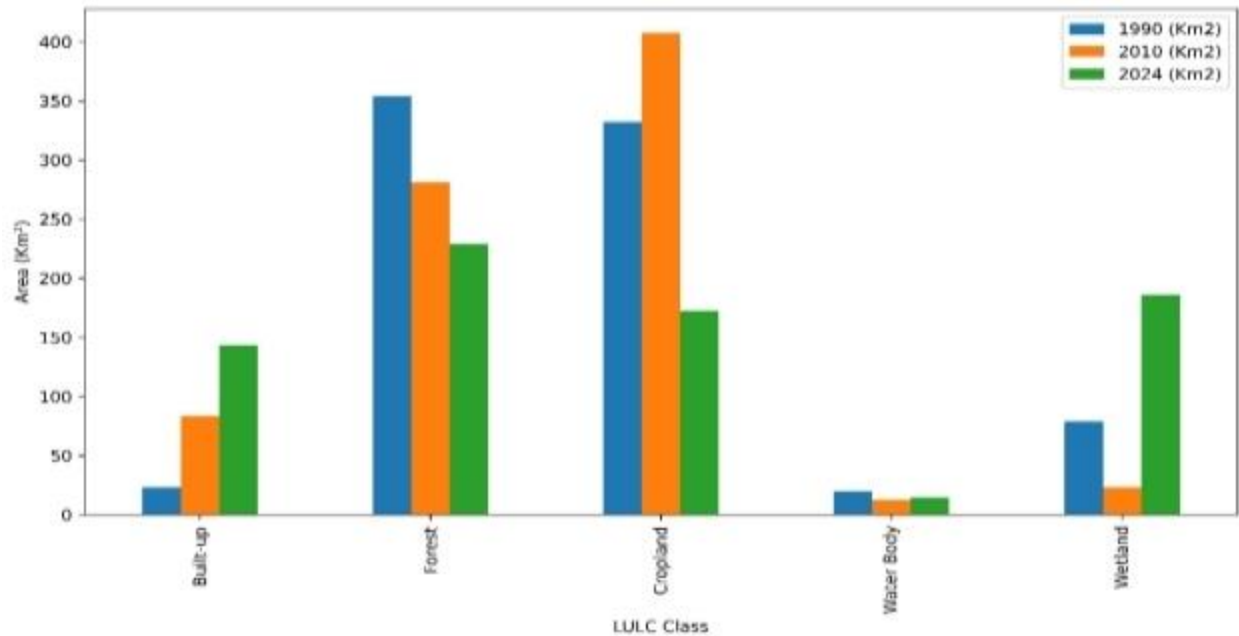


Figure 3.1: Grouped Bar Chart showing the Trend of LULC 1990-2024: Author source

3.1.2 Spatial Interpretation of LULC Maps

The spatial arrangement and evolution of land use/land cover (LULC) classes across the study area reveal clear trajectories of landscape transformation between 1990, 2010 and 2024. The mapped patterns highlight the dynamics of urban growth, vegetation loss, agricultural displacement, and hydrological alteration within the two main urban centres Makurdi and Otukpo and their surroundings as seen in Figure 3.2(a-f).

Spatial Trends in Makurdi

The LULC maps show that Makurdi has undergone a pronounced shift from a predominantly natural and agro-ecological landscape in 1990 to a heavily urbanised environment by 2024. In 1990, built-up areas were limited in extent and spatially concentrated along the central Benue River corridor. The study area was dominated by forest cover, especially in the southern, western, and eastern portions of the region. The grasslands appeared as scattered patches around small settlement clusters, while the wetlands remained restricted to low-lying floodplain zones immediately adjacent to the river.

By 2010, the mapped spatial configuration illustrates significant outward expansion of urban land. In addition, the maps confirmed that the built areas spread laterally along the river corridor toward the northeast and southwest, believed to be driven by infrastructure development and the growth of settlements due to the economic activities along the river corridor. The observed urban expansion in the area coincides specifically with noticeable losses in forest cover, particularly in areas close to transport routes and river adjacent regions. The cropland areas are seen to increase in the northern and central sections where the forests were cleared, while the wetlands showed localized growth resulting from vegetation removal and altered surface drainage systems.

By 2024, urban expansion had become more extensive and spatially dominant. The built-up areas have radiated outward from the central corridor, merging the formerly isolated settlements and also extending deeper into peri-urban zones. The forests experienced continued fragmentation and retreat, now surviving mainly in the far south, southwest, and isolated patches in the east. The maps demonstrated how croplands have become increasingly restricted and how often they are squeezed between expanding built-up areas and remnant forest patches. The wetlands expanded markedly and became more spatially distributed in the central and northern parts, reflecting increased land degradation, runoff concentration, and hydrological disturbance triggered by urban encroachment. In general, the Makurdi landscape exhibits a clear pattern of central to a peripheral urban expansion, which is substantial forest fragmentation, cropland displacement, and widening wetland footprints indicative of increased surface disturbance.

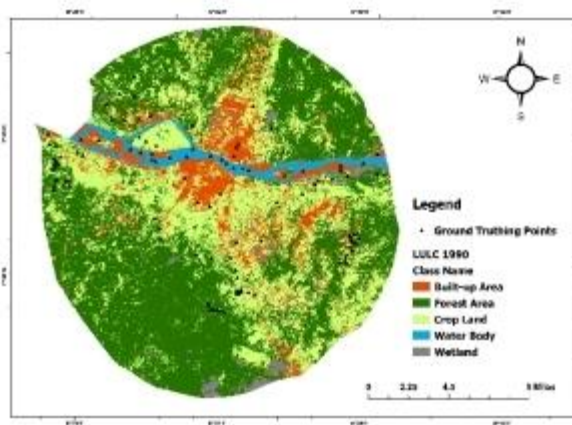
Spatial Trends in Otukpo

In the context of Otukpo, the spatial pattern shows a similar and clear progression by intensifying urban land use throughout the study period. In 1990, the town was characterised by a predominantly natural landscape composed of continuous vegetation and agricultural land. Built-up areas formed only a small, compact core, with minimal encroachment into surrounding rural zones.

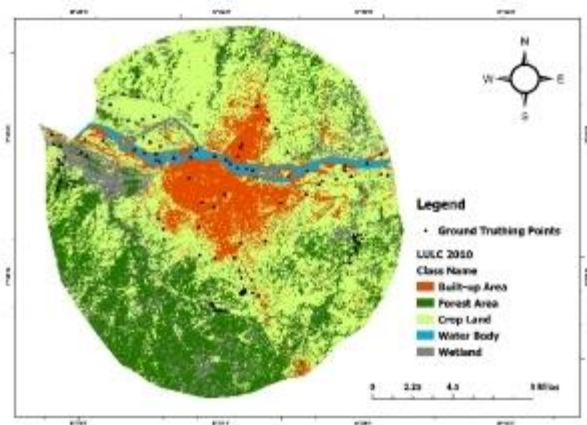
By the 2010 phase, urban development had expanded outside of this original nucleus. The built-up patches spread into peripheral areas and began to fragment in a continuous vegetation and farmland. This period marks the development of a

transitional mixed landscape in which the growth of settlement increasingly influenced the spatial structure of the surrounding environment.

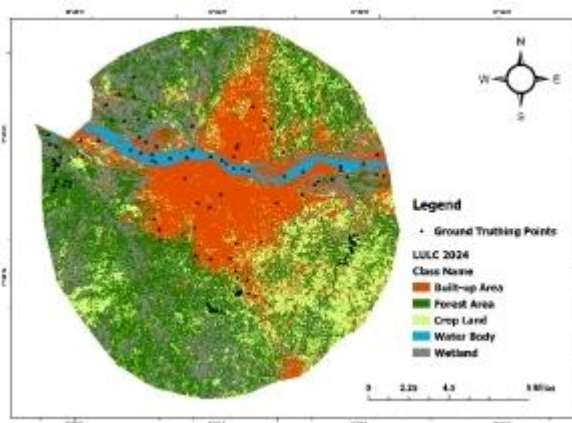
By the 2024 phase, the LULC distribution indicates a significantly enlarged and more spatially continuous urban footprint as the population increases. The built-up areas now dominate the landscape, extending along major roads and settlement corridors that are also drivers of development. This expansion replaced large portions of vegetation and agricultural land, leaving the remaining patches isolated and confined to marginal areas. Vegetation and farmland became highly fragmented and restricted to the outskirts, signalling intensified urban pressure and land conversion.



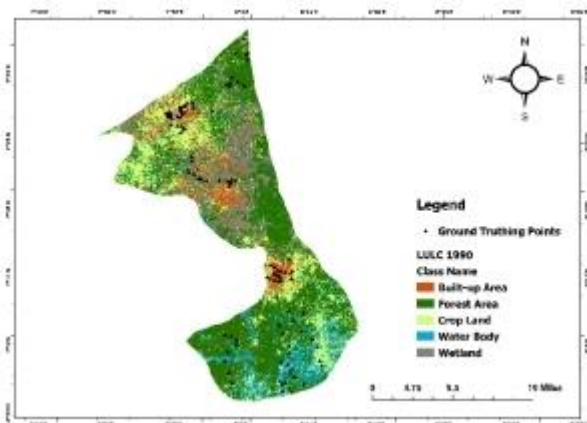
a. Makurdi LULC 1990



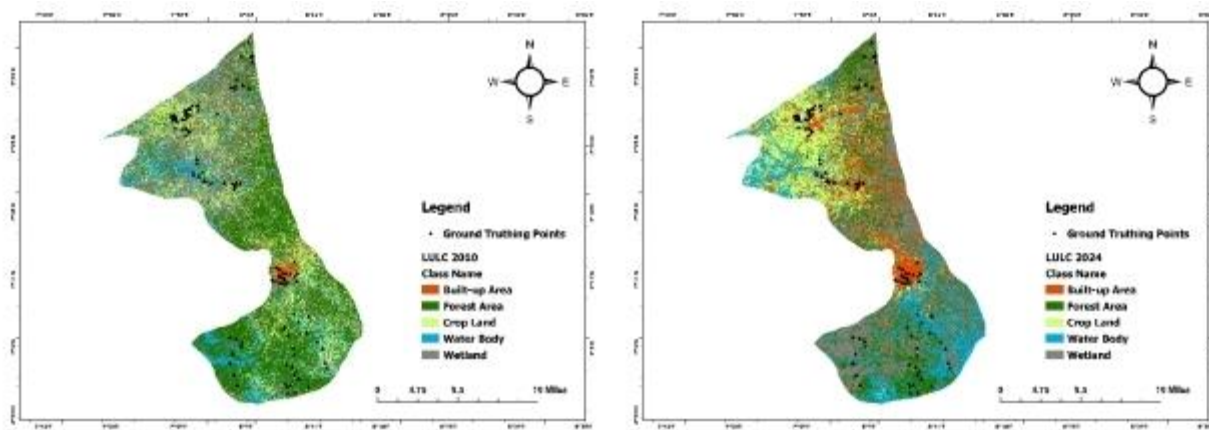
b. Makurdi LULC 2010



c. Makurdi LULC 2024



d. Otukpo LULC 1990



e. Otukpo LULC 2010

f. Otukpo LULC 2024

Figure 3.2: Spatial Trend of LULC for Makurdi and Otukpo between 1990-2024 Author source

3.1.3 Ground Truthing for LULC Classification

An accuracy assessment of land cover classifications was performed to verify the reliability of the results using ground truth data (Appendix 2) from the field survey carried out. Tables 3.2 and Table 3.3 show the results in both study areas

Table 3.2: Accuracy Assessment Result for Makurdi (1990-2024)

LULC Class	Reference Pixels	Correctly Classified	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	45	42	93.3	91.3
Forest Area	60	54	90.0	88.5
Cropland	55	49	89.1	87.5
Water Body	30	29	96.7	96.7
Wetland	40	36	90.0	92.3
Overall Accuracy	—	—	91.8%	—
Kappa Coefficient	—	—	0.89	—

Author source

Table 3.3: Accuracy Assessment Result for Otukpo (1990-2024)

LULC Class	Reference Pixels	Correctly Classified	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	50	47	94.0	90.4
Forest Area	55	49	89.1	89.1
Cropland	45	41	91.1	86.3
Water Body	35	34	97.1	94.4
Wetland	40	36	90.0	90.0
Overall Accuracy	—	—	92.2%	—
Kappa Coefficient	—	—	0.90	—

Author source

3.2 Plant Composition and Biodiversity Patterns

3.2.1 Plant Diversity in Makurdi local Government area

A study was conducted in Makurdi Local Government, encompassing rural, peri-urban, and urban areas, to assess plant diversity. The study covered the following; 1) Agan and Bar wards in rural areas, characterised by agricultural lands, forests, and scattered settlements; 2) Fiidi and Wailomayo wards as peri-urban areas, marked by a blend of developed and natural environments; and 3) North Bank Ward 1 and Ankpa Q/Wadata Ward as urban areas, characterised by high population densities and infrastructure development. By assessing plant diversity across multiple plots in these areas, the study provided valuable information on the distribution and abundance of plant species in different environmental settings.

3.2.1.1 Rural Areas of the Makurdi Local Government

The plant diversity analysis in the Agan Ward disclosed varying levels of diversity across the three plots; see (Appendixes T1-T3). The first plot findings clearly show that 92 plant individuals from 30 species, with *Heterotis rotundifolia*, were found. (10 individuals, 10.9%) and *Elaeis Guineensis* (9 individuals, 9.8%) being notable species. This plot had a Shannon-Wiener index of 3.0431, Simpson's Diversity of 0.9373, and Species Evenness of 0.8947, indicating a relatively high level of

diversity. Plot 2 findings shows that 49 individuals from 15 species, with *Ancardium occidentale* (10 individuals, 20.4%) and *Passiflora foetida* (9 individuals, 18.4%) being prevalent. This plot findings showed lower diversity metrics, with a Shannon-Wiener index of 2.3735, Simpson's Diversity of 0.8821, and Evenness of 0.8765. Plot 3 (Table 3.20) had 35 individuals across 19 species, with *Cocos rufifera* (7 individuals, 20%) and *Magnifera Indica* (4 individuals, 11.4%) being notable. The diversity indices for this plot were 2.7155 (Shannon-Wiener), 0.9143 (Simpson), and 0.9223 (Evenness).

The study of plant diversity in Bar Ward also revealed varying levels of diversity in three plots (Appendixes T4-T6). The first plot recorded 73 individual plants from 25 species, with *Daniellia oliverii* (11 individuals, 15.1%) and *Prosopis africana* (6 individuals, 8.2%) being notable species. This plot had a Shannon-Wiener index of 2.8667, Simpson's Diversity of 0.9266, and Species Evenness of 0.8906. Plot 2 documented 51 individuals from 27 species, with *Talinum fruticosum* (7 individuals, 13.7%) and *Elaeis guinensis* (6 individuals, 11.8%) being prevalent. This plot findings showed higher diversity metrics, with a Shannon-Wiener index of 3.0413, Simpson's Diversity of 0.9373, and Evenness of 0.9228. Plot 3 had 52 individuals across 25 species, with *Magnifera indica* (10 individuals, 19.2%) being notable. The diversity indices for this plot were 2.9171 (Shannon-Wiener), 0.9246 (Simpson), and 0.9062 (Evenness).

3.2.1.2 Peri-Urban Areas of Makurdi Local Government

The plant diversity analysis in Fiidi Ward showed varying levels of diversity across three plots; see (Appendix T7-T9). The first plot findings show that 27 plant individuals from 10 species, with *Eichhornia crassipes*, were present. (4 individuals, 14.8%) and *Alternanthera sessilis* (4 individuals, 14.8%) being notable species. This plot had a Shannon-Wiener index of 2.2428, Simpson's Diversity of 0.8889, and Species Evenness of 0.9740. Plot 2 documented 51 individuals from 15 species, with *Newbouldia laevis* (10 individuals, 19.6%) and *Magnifera indica* (7 individuals, 13.7%) being prevalent. This plot showed a Shannon-Wiener index of 2.4139, Simpson's Diversity of 0.8889, and Evenness of 0.8914. Plot 3 had 31 individuals across 14 species, with *Magnifera indica* (8 individuals, 25.8%) and *Polyalthia longifolia* (4 individuals, 12.9%) being notable. The diversity indices for this plot were 2.3717 (Shannon-Wiener), 0.8783 (Simpson), and 0.8987 (Evenness).

The study of plant diversity in the Wailomayo Ward also revealed varying levels of diversity in three plots (Appendixes T10-T12). The first plot findings show that 57 plant individuals from 20 species, with *Diplazium sammatii* (8 individuals, 14%) and *Albizia zygia* (8 individuals, 14%) being notable species. This plot had a Shannon-Wiener index of 2.7859, Simpson's Diversity of 0.9252, and Species Evenness of 0.9299. Plot 2 findings shows that 23 individuals from 13 species, with *Terminalia mentalis* (6 individuals, 26.1%) being prevalent. This plot showed a Shannon-Wiener index of 2.3440, Simpson's Diversity of 0.8771, and Evenness of 0.9138. Plot 3 findings on the other hand shows that 87 individuals across 23 species, with *Magnifera indica* (10 individuals, 11.5%), *Sida acuta* (11 individuals, 12.6%), and *Cleome viscosa* (9 individuals, 10.3%) being notable. The diversity indices for this plot were 2.8092 (Shannon-Wiener), 0.9256 (Simpson), and 0.8959 (Evenness).

3.2.1.3 Urban Areas of the Makurdi Local Government

The analysis of plant diversity in Ward 1 of the North Bank shows varying levels of diversity in three plots (Appendixes T13-T15). The first plot findings show that 65 plant individuals from 28 species, with *Spermacoce verticellata*, are present. (10 individuals, 15.4%) and *Daniellia oliverii* (5 individuals, 7.7%) being notable species. This plot had a Shannon-Wiener index of 3.0711, Simpson's Diversity of 0.9396, and Species Evenness of 0.9217. Plot 2 documented 58 individuals from 19 species, with *Ixora coccinea* (10 individuals, 17.2%) and *Elaeis guineensis* (9 individuals, 15.5%) being prevalent. This plot showed a Shannon-Wiener index of 2.6457, Simpson's Diversity of 0.9090, and Evenness of 0.8986. Plot 3 had 41 individuals across 25 species, with *Azadrachta indica* (7 individuals, 17.1%) being notable. The diversity indices for this plot were 3.0049 (Shannon-Wiener), 0.9352 (Simpson), and 0.9335 (Evenness).

The plant diversity study in Ankpa Q/Wadata Ward also revealed varying levels of diversity in three plots (Appendixes T16-T18). The first plot findings show that 50 plant individuals from 25 species, with *Cleome viscosa*, were analysed. (10 individuals, 20%) and *Acmela paniculata* (8 individuals, 16%) being notable species. This plot had a Shannon-Wiener index of 2.8483, Simpson's Diversity of 0.9128, and Species Evenness of 0.8849. Plot 2 findings shows that 23 individuals from 9 species, with *Ficus thinningii* (7 individuals, 30.4%) being prevalent. Also the findings shows the plot has a Shannon-Wiener index of 2.0156, Simpson's Diversity

of 0.8393, and Evenness of 0.9173. Plot 3 had 39 individuals across 14 species, with *Delonix regia* (13 individuals, 33.3%) being notable. The diversity indices for this plot were 2.2555 (Shannon-Wiener), 0.8442 (Simpson), and 0.8547 (Evenness).

3.2.1.4 Comparison of the diversity of species between Rural, Peri-Urban and Urban Areas in the Makurdi Local Government

The comparison findings of species diversity between rural, peri-urban, and urban areas in Makurdi Local Government revealed variations in diversity indices, although the differences were not statistically significant (Table 3.4). Rural areas exhibited higher species diversity, with a mean Shannon-Wiener Index (H') of 2.83, Simpson's Diversity (1-D) of 0.92, and Species Richness (R) of 23.50, compared to peri-urban areas ($H' = 2.49$, 1-D = 0.90, R = 15.83) and urban areas ($H' = 2.64$, 1-D = 0.90, R = 20.00). Also, the findings show that the diversity of species (E) was slightly higher in the peri-urban areas (0.90) compared to the rural areas (0.92) and the urban areas (0.90). Despite these differences, the ANOVA results did not show significant differences between areas for any of the diversity indices, indicating that the observed variations may be due to chance.

Table 3.4: Comparison of the species diversity between the Rural, Peri-urban and Urban Areas in the Makurdi Local government of Benue state

Areas	Species Diversity			
	Shannon-Wiener Index (H')	Simpson's Diversity (1-D)	Species Evenness (E)	Species Richness (R)
Rural Areas (Agan and Bar Wards) (n = 6)	2.83 $\pm$ 0.25	0.92 $\pm$ 0.02	0.90 $\pm$ 0.02	23.50 $\pm$ 5.50
Peri-urban Area (Fiidi and Wailomayo wards) (n = 6)	2.49 $\pm$ 0.24	0.90 $\pm$ 0.02	0.92 $\pm$ 0.03	15.83 $\pm$ 4.79
Urban Area (North bank 1 and	2.64 $\pm$ 0.42	0.90 $\pm$ 0.04	0.90 $\pm$ 0.03	20.00 $\pm$ 7.37

Anka wards) (n = 6)	Q/wadata			
F-value	1.644	1.137	0.656	2.463
p-value	0.226	0.347	0.533	0.119

Values are expressed as mean $\pm$ Standard Deviation

Author source

3.2.2 Plant Diversity in Otukpo Local Government

In addition, the study was conducted in Otukpo Local Government, which includes rural, peri-urban, and urban areas, to assess the diversity of plants. Specifically, the study covered the following 1) Adoka-Haje and Ugboju-Eha wards representing rural areas, which are marked by agricultural lands, forests, and scattered settlements; 2) Allan Ward representing the peri-urban area, characterised by a blend of developed and natural environments; and 3) Otukpo Town East and West wards representing the urban areas, which are marked by high population densities and infrastructure development. By assessing plant diversity across multiple plots in these areas, the study provided valuable information on the distribution and abundance of plant species in different environmental settings of the tropical landscape.

3.2.2.2 Peri-Urban Area of Otukpo Local Government

The plant diversity analysis in the Allan Ward shows varying levels of diversity across three plots; see (Appendixes T25-T27). The first plot findings show that 87 individuals of plants from 25 species, with *Heterotis rotundifolia*, were found. (10 individuals, 11.5%), *Malvastrum coromandelianum* (9 individuals, 10.3%), and *Ludwiga abyssininca* (8 individuals, 9.2%) being notable species. This plot had a Shannon-Wiener index of 2.9022, Simpson's Diversity of 0.9322, and Species Evenness of 0.9016. Plot 2 documented 47 individuals from 13 species, with *Ancardium occidentale* (10 individuals, 21.3%) and *Passiflora foetida* (9 individuals, 19.1%) being prevalent. This plot showed a Shannon-Wiener index of 2.2672, Simpson's Diversity of 0.8728, and Evenness of 0.8839. Plot 3 findings shows that 32 individuals across 16 species, with *Cocus nucifera* (7 individuals, 21.9%) and

Magnifera indica (4 individuals, 12.5%) being notable. The findings also indicated that the diversity indices for this plot were 2.4389 (Shannon-Wiener), 0.9014 (Simpson), and 0.8796 (Evenness). These findings suggest that plant diversity varies across plots, with some plots exhibiting higher levels of diversity than others.

3.2.2.1 Rural Areas of the Otukpo Local Government

The plant diversity analysis in Adoka-Haje Ward shows different levels of diversity across three plots; see (Appendix T19-T21). The first plot findings show that 61 plant individuals from 24 species, with *Spermacoce verticellata*, were spotted. (10 individuals, 16.4%) and *Daniellia oliverii* (5 individuals, 8.2%) being notable species. This plot had a Shannon-Wiener index of 2.9353, Simpson's Diversity of 0.9325, and Species Evenness of 0.9236. Plot 2 documented 56 individuals from 17 species, with *Ixora coccinea* (10 individuals, 17.9%) and *Elaeis guineensis* (9 individuals, 16.1%) being prevalent. This plot showed a Shannon-Wiener index of 2.5601, Simpson's Diversity of 0.9031, and Evenness of 0.9036. Plot 3 had 36 individuals across 20 species, with *Azadrachta indica* (7 individuals, 19.4%) being notable. The diversity indices for this plot were 2.7765 (Shannon-Wiener), 0.9198 (Simpson), and 0.9268 (Evenness).

The study of plant diversity in the Ugboju-Eha Ward also revealed varying levels of diversity in three plots (Appendixes T22-T24). The first plot findings show that 63 plant individuals from 20 species, with *Cleome viscosa*, are present. (10 individuals, 15.9%) and *Acmela paniculata* (8 individuals, 12.9%) being notable species. This plot had a Shannon-Wiener index of 2.7017, Simpson's Diversity of 0.9161, and Species Evenness of 0.9019. Plot 2 findings shows that 44 individuals from 17 species, with *Ficus thinningii* (7 individuals, 15.9%) being prevalent. This findings also shows that the plot have a Shannon-Wiener index of 2.6670, Simpson's Diversity of 0.9205, and Evenness of 0.9449. Plot 3 had 59 individuals across 22 species, with *Delonix regia* (13 individuals, 22.0%) being notable. The diversity indices for this plot were 2.7677 (Shannon-Wiener), 0.9112 (Simpson), and 0.8954 (Evenness).

3.2.2.3 Urban Areas of the Otukpo Local Government

The plant diversity analysis in Otukpo Town East Ward showed varying levels of diversity across three plots, see (Appendix T28-T30). The first plot findings show

that 55 plant individuals from 18 species, with *Diplazium sammatii*, are present. (8 individuals, 14.5%), *Albizia zygia* (8 individuals, 14.5%), and *Eichhornia crassipes* (5 individuals, 9.1%) being notable species. This plot had a Shannon-Wiener index of 2.7044, Simpson's Diversity of 0.9203, and Species Evenness of 0.9357. Plot 2 documented 40 individuals from 19 species, with *Tephrosia linearis* (8 individuals, 20.0%) being prevalent. This plot showed a Shannon-Wiener index of 2.6923, Simpson's Diversity of 0.9125, and Evenness of 0.9144. Plot 3 findings shows that 84 individuals across 20 species, with *Sida acuta* (11 individuals, 13.1%), *Magnifera indica* (10 individuals, 11.9%), and *Cleome viscosa* (9 individuals, 10.7%) being notable. The diversity indices for this plot were 2.7149 (Shannon-Wiener), 0.9206 (Simpson), and 0.9063 (Evenness).

The study of plant diversity in the West Ward of Otukpo also revealed varying levels of diversity in three plots (Appendixes T31-T33). The first plot findings show that 25 plant individuals from 9 species, with *Eichhornia crassipes*, were present. (4 individuals, 16%) and *Alternanthera sessilis* (4 individuals, 16%) being notable species. This plot had a Shannon-Wiener index of 2.1370, Simpson's Diversity of 0.8768, and Species Evenness of 0.9726. Plot 2 documented 47 individuals from 13 species, with *Newbouldia laevis* (10 individuals, 21.3%) and *Magnifera indica* (7 individuals, 14.9%) being prevalent. This plot showed a Shannon-Wiener index of 2.2732, Simpson's Diversity of 0.8737, and Evenness of 0.8862. Plot 3 had 42 individuals across 16 species, with *Piliostigma thinningii* (5 individuals, 11.9%) and *Khaya senegalensis* (5 individuals, 11.9%) being notable. The diversity indices for this plot were 2.6239 (Shannon-Wiener), 0.9195 (Simpson), and 0.9464 (Evenness).

3.2.2.4 Comparison of the diversity of species between Rural, Peri-Urban and Urban Areas in the Otukpo Local Government

The study compared the diversity of species in rural, peri-urban and urban areas in the Otukpo Local Government (Table 3.5). The results showed a slightly higher species diversity in rural areas compared to peri-urban and urban areas, with rural areas having a mean Shannon-Wiener Index (H') of 2.74, while species diversity was similar between peri-urban and urban areas, both having mean Shannon-Wiener Index (H') values of 2.54 and 2.52, respectively. The findings also show that Simpson's diversity ($1-D$) was similar in all areas, ranging from 0.90 to 0.92. Species Evenness (E) was also similar, with values ranging from 0.89 to 0.93. Species

Richness (R) was higher in rural and peri-urban areas (20.00 and 18.00, respectively) compared to urban areas (15.83).). However, the ANOVA results showed no significant differences between the areas for any of the diversity indices, indicating that the observed differences may be due to chance.

Table 3.5: Comparison of species diversity between the Rural, urban, and urban areas of Peri in the Otukpo Local government of Benue state: Author source

Areas	Species Diversity			
	Shannon-Wiener Index (H')	Simpson's Diversity (1-D)	Species Evenness (E)	Species Richness (R)
Rural Areas (Adoka-Haje and Ugboju-Eha Wards) (n = 6)	2.74 $\pm$ 0.12	0.92 $\pm$ 0.01	0.92 $\pm$ 0.02	20.00 $\pm$ 2.76
Periurban Area (Allan ward) (n = 3)	2.54 $\pm$ 0.33	0.90 $\pm$ 0.03	0.89 $\pm$ 0.01	18.00 $\pm$ 6.24
Urban Area (Otukpo City East and Otukpo City West wards) (n = 6)	2.52 $\pm$ 0.25	0.90 $\pm$ 0.02	0.93 $\pm$ 0.03	15.83 $\pm$ 4.17
F-value	1.528	0.882	2.621	1.541
p-value	0.256	0.439	0.114	0.254

Values are expressed as mean $\pm$ Standard Deviation

3.3 Soil Properties on the Urban Gradient

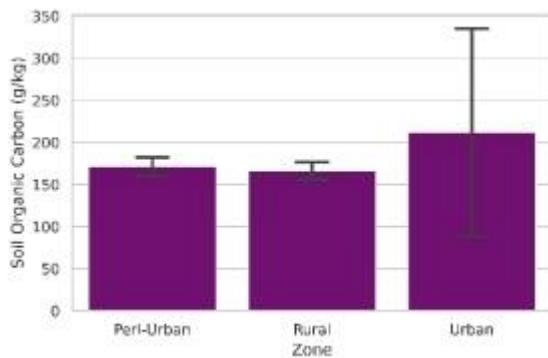
3.3.1 Summary of soil variables

The finding of the analysis shows that soil properties in urbanization gradients exhibit clear patterns reflecting the influence of urbanization. Organic carbon and

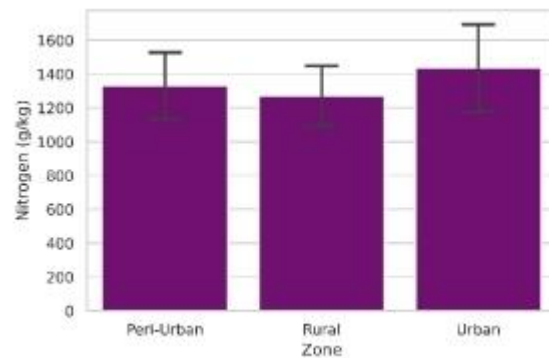
nitrogen increase from rural to urban areas, likely as a result of higher organic inputs and intensified human activity. The findings further show that the bulk density remains relatively uniform across the zones, while the soil pH gradually rises from peri-urban to urban areas. The cation exchange capacity is highest in urban and peri-urban zones, with lower values observed in rural areas. These results indicate that soil characteristics respond strongly to the level of urban development. Table 3.6 summarizes key soil indicators and Figure 3.3(a–e) illustrates how each variable varies along the urbanization gradient.

Table 3.6: Summarized Reporting Table: Author source

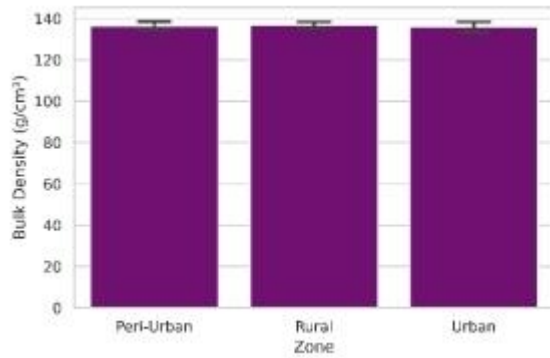
Zone	SOC (mean)	N (mean)	Bulk Density (mean)	pH (mean)	CEC (mean)
Peri-Urban	171.18	1331.15	136.38	56.82	100.30
Rural	166.27	1269.96	136.68	57.05	88.51
Urban	211.99	1435.58	135.87	58.87	102.57



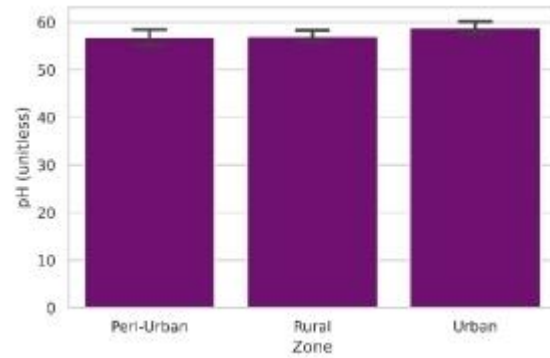
a. SOC Trend



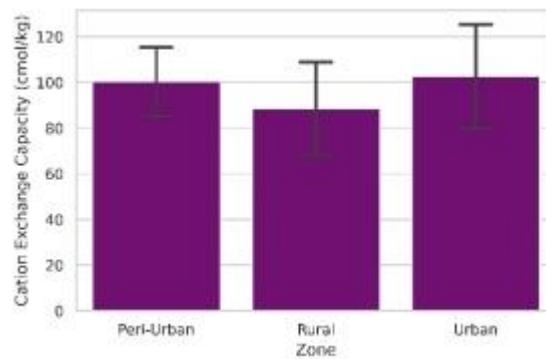
b. N Trend



c. BD Trend



d. pH Trend



e. CEC Trend

Figure 3.3: Trends of soil variables in urban gradients: Author source

3.3.2 Statistical Differences between Zones

The ANOVA findings shown in Table 3.7 show that among the five soil variables measured, only pH differed significantly between zones ($F = 4.061$, $p = 0.037$, $\eta^2 = 0.337$). Post-hoc comparisons (Table 3.8) reveal that this significant difference occurs between the Peri-Urban and Urban zones. The findings further show that other soil properties, including SOC, N, BD and CEC, did not show statistically significant variation between zones. In general, these findings suggest that urbanization had a limited impact on soil chemical and physical properties, except soil pH, which may reflect localised anthropogenic influences.

Table 3.7: ANOVA Results and Effect Sizes for Soil Variables Across Zones: Author source

Variable	F-value	p-value	η^2 (Effect Size)	Significance
SOC	0.799	0.467	0.091	NS
N	0.940	0.411	0.105	NS
BD	0.235	0.793	0.029	NS
pH	4.061	0.037	0.337	<i>Significant</i>
CEC	0.923	0.418	0.103	NS

Table 3.8: Tukey Post-hoc Comparisons Between Zones for Soil Variables: Author source

Variable	Comparison	Mean Difference	p-adjusted	Significant
SOC	Peri-Urban vs Rural	-4.91	0.991	No
SOC	Peri-Urban vs Urban	40.81	0.553	No
SOC	Rural vs Urban	45.71	0.504	No
N	Peri-Urban vs Rural	-61.19	0.863	No
N	Peri-Urban vs Urban	104.43	0.657	No
N	Rural vs Urban	165.62	0.387	No
BD	Peri-Urban vs Rural	0.297	0.964	No
BD	Peri-Urban vs Urban	-0.512	0.898	No
BD	Rural vs Urban	-0.809	0.780	No
pH	Peri-Urban vs Rural	0.224	0.955	No
pH	Peri-Urban vs Urban	2.050	0.044	Yes
pH	Rural vs Urban	1.826	0.089	No
CEC	Peri-Urban vs Rural	-11.79	0.532	No
CEC	Peri-Urban vs Urban	2.28	0.976	No
CEC	Rural vs Urban	14.06	0.439	No

3.4 Vegetation Index (NDVI/EVI) Results

3.4.1 Summary of NDVI and EVI by Zone

Vegetation dynamics across the urban gradient was assessed using the NDVI and the EVI. Mean values and standard deviations were calculated for each zone (Rural, Peri-Urban, and Urban) to evaluate the response of vegetation cover to urbanization pressures; see Table 3.9 and Figure 3.4.

The findings show that rural areas exhibit the highest vegetation indices, with a mean NDVI of 0.48 and a mean EVI of 0.31, reflecting relatively dense and healthy vegetation cover. In addition, the Peri-urban gradients show intermediate values of (NDVI = 0.43, EVI = 0.28), suggesting partial modification of the landscape due to human activity. Urban zones have the lowest indices (NDVI = 0.28, EVI = 0.17), highlighting the reduction in vegetation cover associated with intensive urbanization.

Table 3.9: Mean NDVI and EVI in Zones: Author source

Zone	NDVI (mean $\pm$ SD)	EVI (mean $\pm$ SD)
Peri-Urban	0.427 $\pm$ 0.027	0.283 $\pm$ 0.012
Rural	0.482 $\pm$ 0.023	0.312 $\pm$ 0.025
Urban	0.277 $\pm$ 0.076	0.173 $\pm$ 0.051

Note: NDVI and EVI are unitless indices ranging from 0 to 1, where higher values indicate higher vegetation density and greenness.

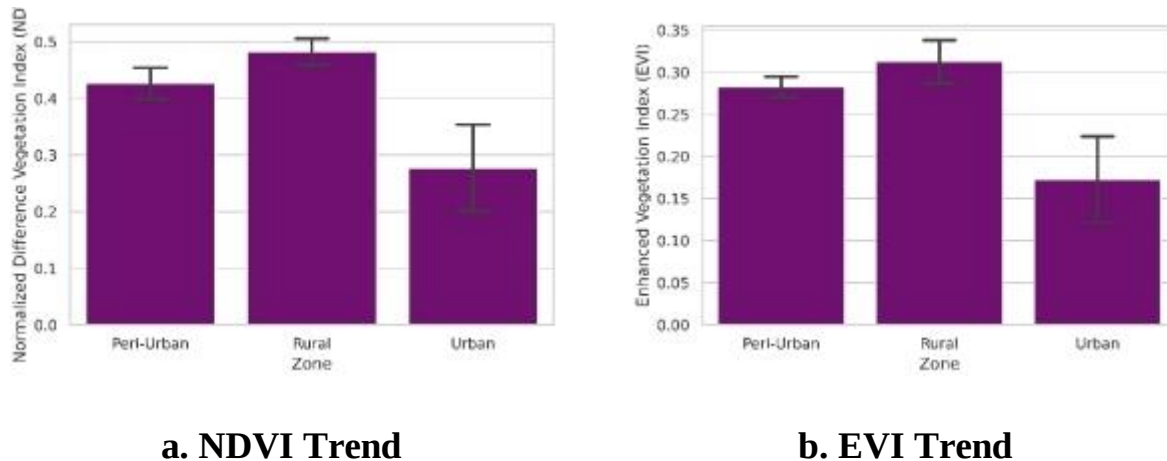


Figure 3.4: Bar chart showing mean NDVI and EVI in Urban Zones: Source of the author

The bar graph illustrates the decline in vegetation indices from rural to urban zones, visually reinforcing the quantitative trends shown in Table 3.9. NDVI and EVI both decrease as the built-up intensity increases, highlighting the impact of urbanization on ecosystem functioning.

3.4.2 Zone Comparison Statistics

To assess whether vegetation indices differ significantly across the urban gradient, one-way ANOVA tests were conducted for NDVI and EVI see Table 3.10. The findings show that there are highly significant differences between zones for both indices (NDVI: $F = 31.04$, $p < 0.001$, $\eta^2 = 0.75$; EVI: $F = 32.52$, $p < 0.001$, $\eta^2 = 0.76$), demonstrating that urbanization has a substantial effect on vegetation cover and greenness.

Table 3.10: ANOVA Results for NDVI and EVI in zones: Author source

Variable	F-value	p-value	η^2
NDVI	31.04	5.35×10^{-7}	0.747
EVI	32.52	3.71×10^{-7}	0.756

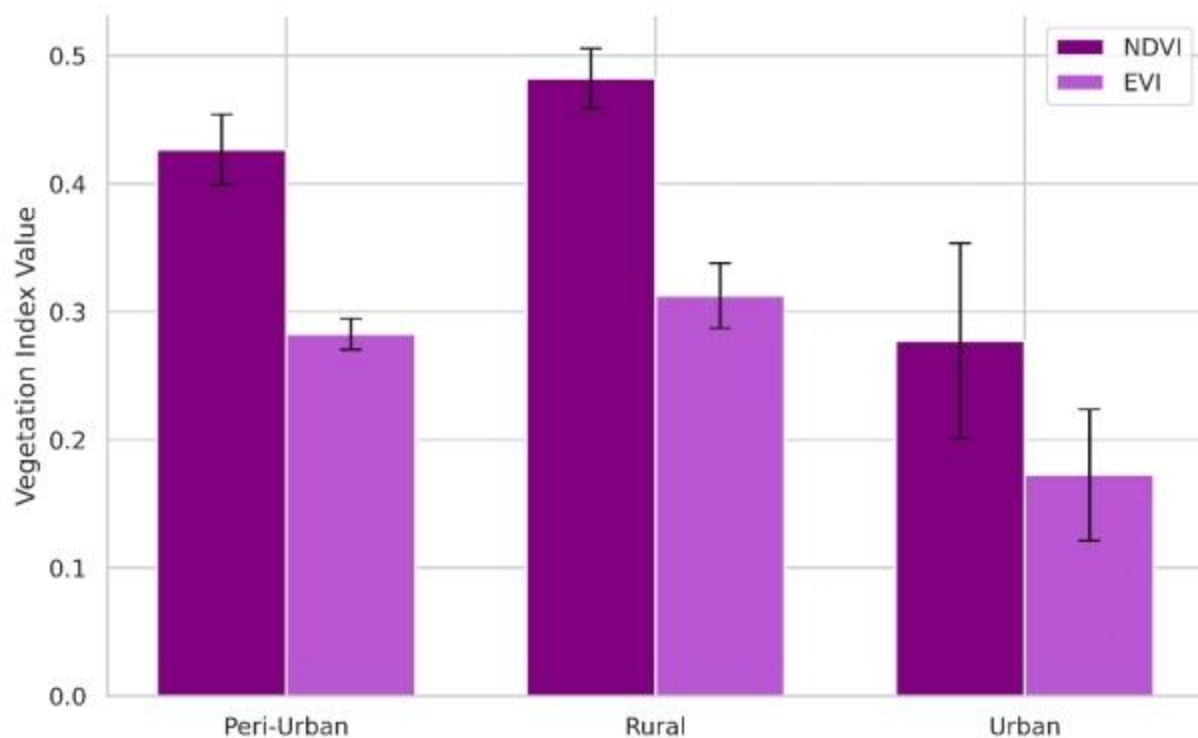
Post-hoc pairwise comparisons using Tukey's test further highlighted the significant differences between specific zone pairs, see Table 3.11. The NDVI and EVI values

in urban areas were significantly lower than in rural zones, while the differences between peri-urban and rural areas were not always significant.

Table 3.11: Tukey Post-hoc Comparisons for NDVI and EVI: Author source

Variable	Comparison	Mean Difference	p-adjusted	Significant
NDVI	Peri-Urban vs Rural	0.056	0.1941	No
NDVI	Peri-Urban vs Urban	-0.149	0.000	Yes
NDVI	Rural vs Urban	-0.205	0.000	Yes
EVI	Peri-Urban vs Rural	0.030	0.3512	No
EVI	Peri-Urban vs Urban	-0.110	0.000	Yes
EVI	Rural vs Urban	-0.140	0.000	Yes

These results confirm that urbanization significantly reduces vegetation cover and greenness, which is also seen in Figure 3.5. NDVI and EVI decline progressively from rural to urban zones, with the largest reductions observed in highly developed urban areas.



The figure visually illustrates the sharp decrease in the vegetation indices from rural to urban areas, supporting the statistical evidence presented in Tables 3.10 and 3.11.

Figure 3.5: Bar chart showing the differences in NDVI and EVI between zones.: Author source

3.5 Relationship Between Urbanization Metrics and Ecosystem Functioning

3.5.1 Summary of Urbanization Indicators

To establish the basic outline of urban growth across the study wards in both LGAs, there are three key indicators that were summarised: 1) built-up intensity, 2) night-time lights (NTL) and 3) population density (POP). Table 3.12 presents the descriptive statistics for these variables.

The percentage of build-up (intensity) had an average value of 2.64, with values ranging from 2.00 to 5.68, reflecting the expected gradient from rural to highly urbanised wards. The night lights showed a mean of 1.69, but with a relatively large standard deviation (1.69) and a wide spread (0.26–5.68), consistent with strong

contrasts in electrification and economic activity within the landscape. The findings also show that population density also exhibited substantial variability, with values ranging from 0.64 to 59.13 and an overall mean of 15.36, signalling the coexistence of sparsely settled peri-urban/rural wards and densely populated urban centres.

Table 3.12: Descriptive Statistics for Urbanization Variables: Author source

Variable	n	Mean	Std Dev	Min	Median	Max
Built-Up	24	2.64	1.02	2.00	2.01	5.68
NTL	24	1.69	1.69	0.26	0.73	5.68
POP	24	15.36	17.51	0.64	3.88	59.13

To explore the interrelationships among the three indicators, the Pearson correlation coefficients were calculated (Table 3.13 and Figure 3.6). All three-urbanization metrics were strongly and positively correlated with each other. Built-up intensity was very highly correlated with both NTL ($r = 0.84$) and population density ($r = 0.91$). Similarly, NTL and population density were also strongly correlated ($r = 0.88$).

The corresponding p-values (Table 3.14) were all statistically significant ($p < 0.001$), confirming that these associations are unlikely to arise by chance. These patterns indicate that the urbanization indicators are closely interlinked, that is, wards with higher population density tend to have more built-up surfaces and stronger night-time illumination. This multicollinearity has diagnostic implications for regression modelling and justifies the use of variance inflation factor (VIF) checks in Section 3.5.3.

Table 3.13: Correlation Matrix for Urbanization Variables

	BuiltUP	NTL	POP
BuiltUP	1.00	0.84	0.91
NTL	0.84	1.00	0.88
POP	0.91	0.88	1.00

Author source

Table 3.14: p-values for Correlations

	BuiltUP	NTL	POP
BuiltUP	0	3.10×10^{-7}	5.66×10^{-10}
NTL	3.10×10^{-7}	0	1.31×10^{-8}
POP	5.66×10^{-10}	1.31×10^{-8}	0

Author source

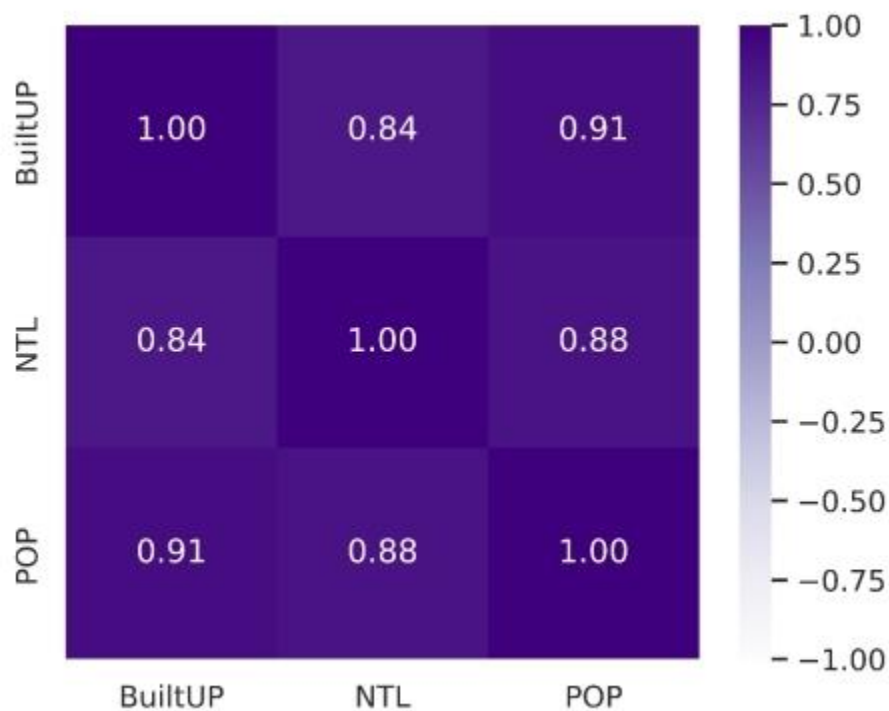


Figure 3.6: Heatmap showing the correlation between Urbanization Variable: Author source

3.5.2 Regression Analysis of Urbanization Metrics as Predictors of NDVI

To quantify how urbanization pressures influence vegetation greenness, a multiple linear regression model was fitted using 1) Built-up index, 2) night-time lights (NTL) and 3) population density (POP) as predictors of NDVI in the 24 wards

sampled. The model performed strongly, indicating that a considerable proportion of the spatial variation in NDVI is attributable to these three urbanization indicators.

The overall model explained 86.2% of the variation in NDVI ($R^2 = 0.862$), with an adjusted R^2 of 0.842, confirming high explanatory power even after adjusting for the number of predictors (Table 3.15). The model’s F-statistic was significant showing ($F = 41.73$, $p < 0.001$), proving that the predictor set collectively contributes meaningfully to NDVI variation. Information criteria ($AIC = -80.28$; $BIC = -75.57$) also support a well-fitted model.

A closer evaluation of the individual coefficients showed contrasting effects (Table 3.16).

The built-up has a predicator that had a positive and statistically significant effect ($\beta = 0.0537$, $p = 0.020$), suggesting that more built-up areas in these zones are associated with small but notable increases in NDVI likely reflecting mixed vegetation–settlement mosaics in peri-urban landscapes.

NTL exhibited a non-significant negative influence ($\beta = -0.0130$, $p = 0.258$), implying that light intensity alone is not a strong predictor of vegetation greenness.

However, population density had a strong negative and highly significant effect ($\beta = -0.0072$, $p < 0.001$), indicating that areas with greater human population pressure tend to show lower NDVI, consistent with clearing, disturbance and land conversion.

Residual diagnostics (Appendix X) did not reveal extreme deviations, suggesting that the model fits the data well in rural, peri-urban, and urban contexts.

Table 3.15: Model Performance Statistics for the NDVI Regression Model: Author source

Metric	Value
R^2	0.8623
Adjusted R^2	0.8416
F-Statistic	41.7327

p-value (F)	8.51×10^{-9}
AIC	−80.28
BIC	−75.57
n	24

Table 3.16: Regression Coefficients for Built-Up, NTL, and Population Predicting NDVI

Term	Coefficient	Std. Error	t-Statistic	p-value	95% CI (Lower)	95% CI (Upper)
Constant	0.3630	0.0400	9.0645	<0.001	0.2795	0.4466
Built-up	0.0537	0.0213	2.5189	0.020	0.0092	0.0982
NTL	−0.0130	0.0112	−1.1640	0.258	−0.0364	0.0103
Population	−0.0072	0.0014	−5.0307	<0.001	−0.0102	−0.0042

Author source

The regression results show that the variability of NDVI in the study areas is strongly shaped by the underlying intensity of urbanization. Built-up extent exerts a mixed but significant influence, possibly due to transitional landscapes where vegetation remains intersperse The Built-up development. In contrast, population density is a robust suppressor of vegetation greenness, consistent with urban expansion, land conversion, and reduced green cover in densely populated wards. In conclusion, these patterns reflect broader findings that urban pressure indicators tend to significantly reduce vegetation productivity in rapidly urbanizing cities like the study area.

3.5.3 Model Fit Validation

A comprehensive and standard diagnostic evaluation was conducted to assess whether the multiple linear regression model that links the NDVI to urbanisation indicators satisfied the primary assumptions required for inferential reliability. Diagnostics included assessments of multicollinearity, residual normality, homoscedasticity, and influence statistics. Also, a graphical diagnostics, namely the

QQ plot of standardized residuals and the Residuals-versus-Fitted plot, were also examined for visual confirmation of model adequacy. Together, these procedures ensure that the estimated regression parameters can be interpreted with statistical confidence.

Multicollinearity Assessment (Variance Inflation Factor)

Multicollinearity among the predictors was evaluated using the Variance Inflation Factor (VIF). Table 3.17 shows that all VIF values are below the conventional threshold of 10, indicating the absence of severe multicollinearity. Where the NTL and the population density show relatively higher VIFs (9.44 and 8.65), they remain within acceptable limits, suggesting that the predictors contribute unique explanatory information to the model.

Table 3.17: Variance Inflation Factors for Urbanization Predictors

Variable	VIF
BuiltUP	4.63
NTL	9.44
POP	8.65

Author source

Residual Normality Assessment

Residual normality was evaluated using both visual and statistical approaches.

Visual Assessment (QQ Plot)

Figure 3.7 shows the QQ plot that compares the standardized residuals with the theoretical normal distribution of the analysis. Where the points align closely with the 1:1 reference line, with minor deviations at the upper and lower tails. This visual pattern indicates that the residuals approximate normality and that the model errors are not characterized by heavy tails or systematic departures from Gaussian behavior.

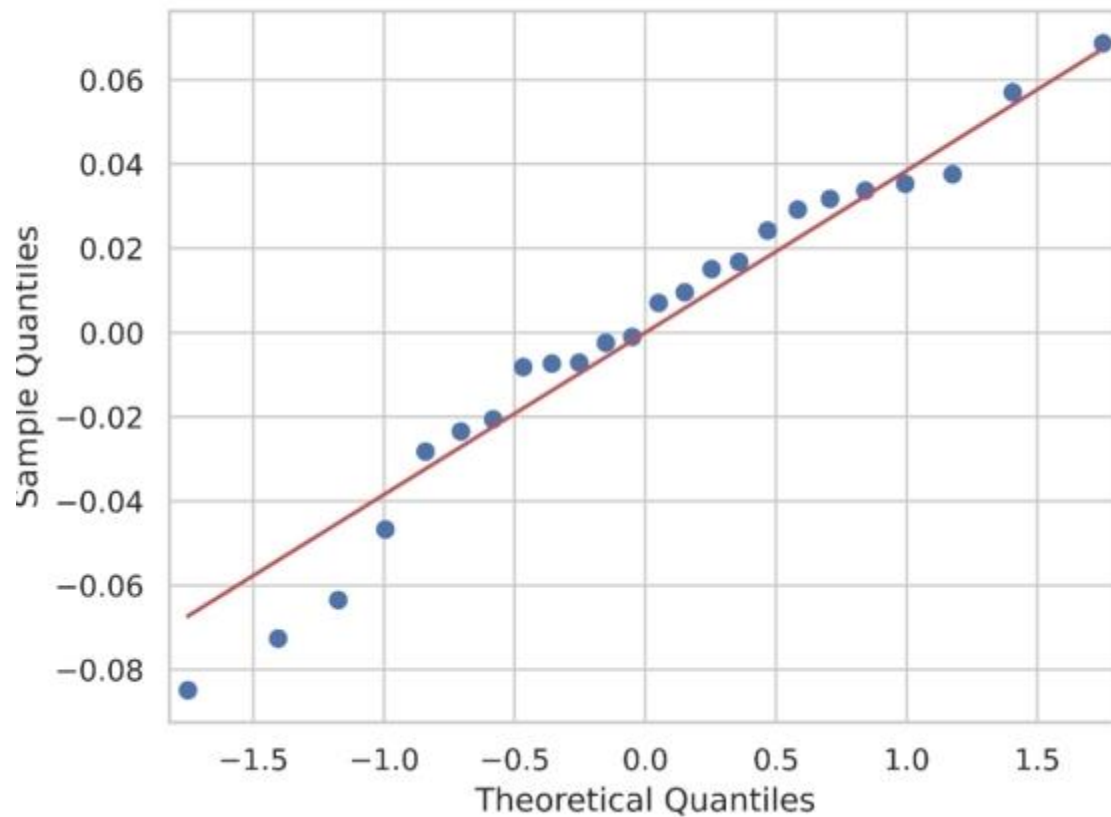


Figure 3.7: QQ plot showing standardized residuals compared with the theoretical normal distribution.: Author source

Shapiro-Wilk Normality Tests

In addition to the visual diagnosis, the Shapiro-Wilk tests were conducted for each zone and for all combined wards. Tables 3.18 and 3.19 show that all p-values exceeded 0.05, indicating that there was no statistically significant deviation from normality between zones.

Table 3.18: Shapiro-Wilk Normality Test by Zone

Zone	W	p_value	n
Peri-Urban	0.9642	0.854	7
Rural	0.9588	0.811	6
Urban	0.9373	0.490	11

Author source

Table 3.19: Overall Shapiro–Wilk Normality Test

Zone	W	p_value	n
All Wards	0.9653	0.554	24

Author source

These results confirm that the assumption of normally distributed residuals is satisfied.

Homoscedasticity Assessment

Homoscedasticity the assumption of constant residual variance was assessed using the Breusch-Pagan test and visual inspection of the Residuals-versus-Fitted plot.

Breusch-Pagan Test

The Breusch-Pagan test returned a non-significant Lagrange multiplier statistic (LM = 3.747, $p = 0.290$), as shown in Table 3.20, indicating that heteroskedasticity is not present in the model.

Table 3.20: Breusch-Pagan Test for Heteroscedasticity

LM statistic	p-value	F-value	F p-value
3.747	0.290	1.234	0.324

Visual Assessment (Residuals vs Fitted Plot)

Figure 3.8 showed the standardized residuals plotted against the fitted NDVI values. in which the random dispersion of points around the zero line, without funnel patterns or curvature, confirms homoscedasticity and validates the linearity assumption. The findings of the analysis also show that the model performs consistently across wards of varying urbanisation intensities.

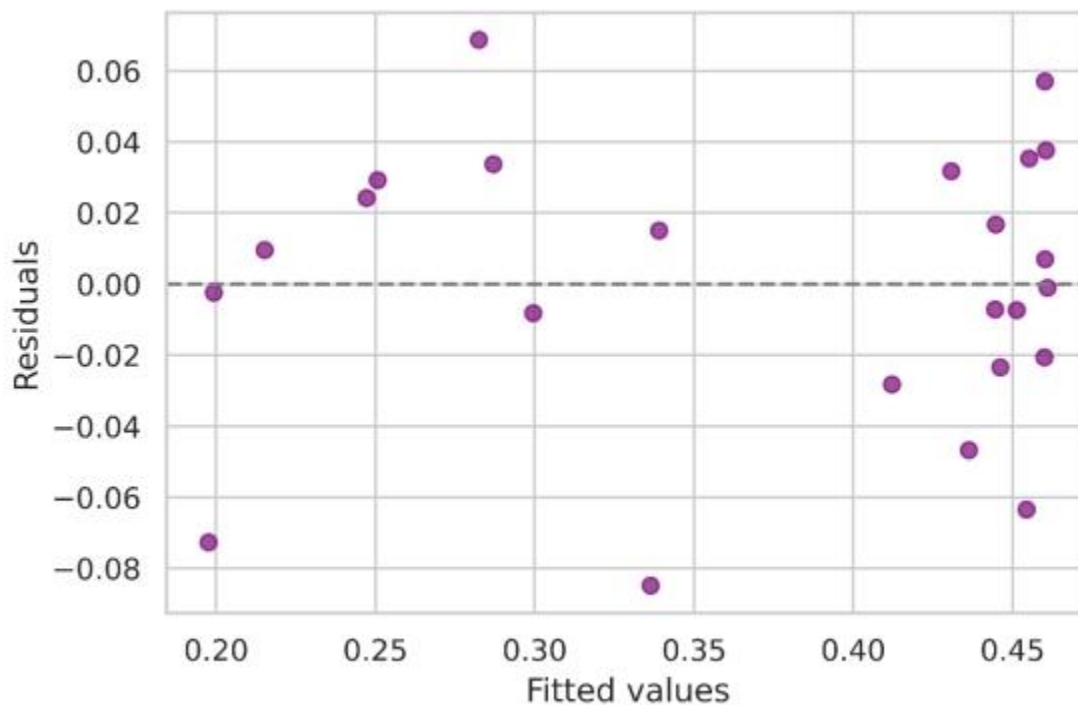


Figure 3.8: Residuals-versus-fitted plot assessing homoscedasticity of model errors: Author source

Influence Diagnostics (Cook's Distance)

Influential observations were assessed through the Cook's distance and the findings as shown in Figure 3.9 showed the Cook's distance values for all wards, while Table 3.21 lists the wards with the highest influence scores. Most of the wards exhibited very low Cook's distances, indicating limited leverage on model estimates. From the findings, two wards Clerks/Market (Urban) and Otukpo East (Urban) showed a

comparatively higher influence, with Cook’s distance values of 2.001 and 0.251, respectively. These values suggest that, while the model is generally stable, these wards merit consideration during interpretation.

Table 3.21: Cook’s Distance for Influential Wards: Author source

Ward	Zone	cooks_d
Clerks/Market	Urban	2.001
Otukpo East	Urban	0.251

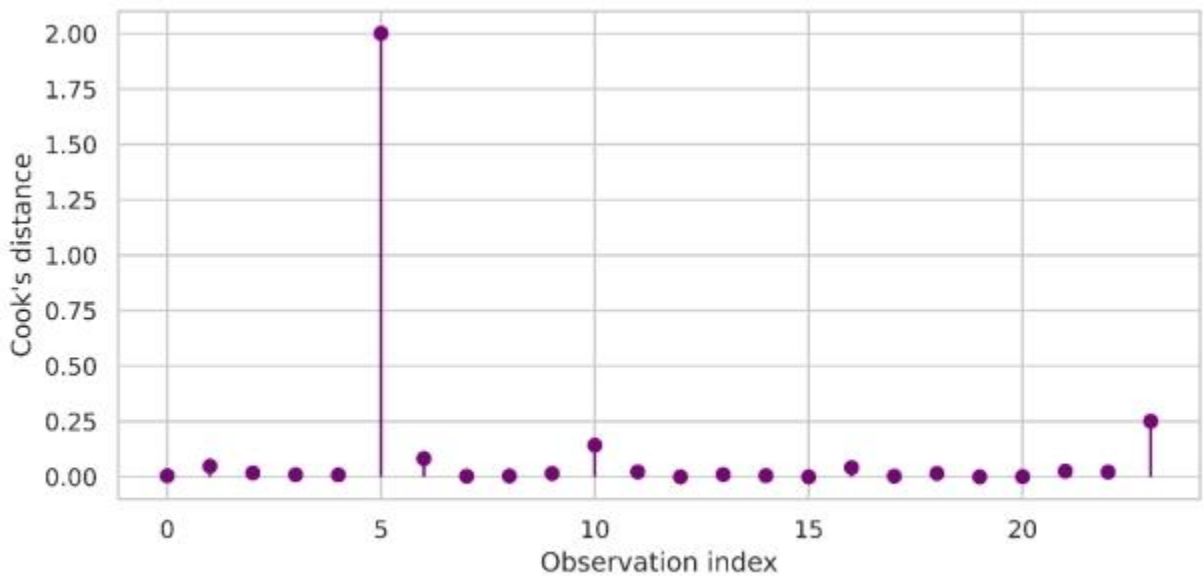


Figure 3.9: Cook’s distance values for all wards, indicating possible influential observations: author source

The overall findings show that the regression model satisfies the primary statistical assumptions required for reliable inference. Multicollinearity among the predictors is minimal, with all the values of the variance inflation factor (VIF) falling below the critical threshold of 10, indicating acceptable and independences between the urbanization indicators considered for this study. Residual analyses further show that

the errors follow an approximately normal distribution, as evidenced by the alignment of points along the QQ plot as well as the non-significant Shapiro-Wilk test results across all zones and for the pooled dataset. The Homoscedasticity was also maintained, which was supported both by the random scatter pattern observed in the residuals-versus-the fitted values plot and by the non-significant Breusch-Pagan test, which indicates constant error variance across fitted values. The Influence diagnostics based on Cook's distance also shows that the model is generally stable, with only two wards exerting modest influence on the parameter estimates. However, collectively, these diagnostic evaluation findings confirm that the regression model is statistically robust and that the coefficient estimates can be interpreted with confidence.

3.5.4 Interpretation of Regression Results

The regression model offers important information on how different dimensions of urbanization built-up extent, night-time lights, and population density shape vegetation condition (NDVI) in study areas. The direction and magnitude of the coefficients, together with their statistical significance, reveal distinct pathways through which urban development influences ecological functioning.

Direction and Strength of Predictor Impacts

Built-up as a variable in this study, shows a positive and statistically significant influence on NDVI ($\beta = 0.0537$, $p = 0.020$). Although built-up expansion is commonly associated with vegetation loss, the positive direction observed here suggests that in the study area, built-up surfaces are frequently embedded within mixed settlement-vegetation mosaics, especially in peri-urban zones. The pattern shows how small-scale cultivation, the household tree stands, the fallow vegetation and the compound greenery coexist within developing neighbourhoods, which contributes to slightly elevated NDVI values even as the built structures increase.

NTL shows a negative but non-significant coefficient ($\beta = -0.0130$, $p = 0.258$), and this implied that illumination intensity alone is not a strong or direct determinant of vegetation status. This weak relationship is consistent with the role of NTL as a proxy for economic activity or electrification rather than transformation of land cover. Although brighter areas tend to be more urbanised, NTL does not

independently account for variations in vegetation greenness once population and built-up surfaces are considered.

The density of population shows the strongest and most significant negative effect on NDVI as ($\beta = -0.0072$, $p < 0.001$), reflected the substantial ecological pressure exerted by the concentrated presence of humans. A higher population density commonly induces vegetation clearance for housing, commercial activity, fuelwood extraction, transportation corridors, and industrial development. This makes population density a robust suppressor of ecosystem productivity within rapidly urbanizing environments.

Comparative Importance of Predictors

Population density stands out in terms of relative strength as the most influential variable in the model, both statistically and ecologically. Its large negative coefficient and high t-value (-5.03) demonstrate that demographic pressure is the dominant mechanism through which urbanization degrades vegetation. However, the built-up area, although significant, has a comparatively modest effect and is consistent with transitional landscapes where vegetation persists within developing communities. And then, on the contrary, the weak signal of NTL underscores that electrification or economic intensity does not directly translate into vegetation decline unless it is accompanied by population-driven land conversion.

Implications for Urban Ecological Dynamics

Taking into account all, the regression findings showed that although urbanisation indicators are strongly correlated, they do not contribute equally to ecological outcomes. Similarly, the urbanisation zones expressed in the descriptive statistics spanning from rural, peri-urban, and urban wards translate into a clear ecological gradient in which population pressure is the principal driver of vegetation depletion.

The regression model therefore highlights three key ecological insights:

1. That urban expansion without high population concentration may retain pockets of vegetation, explaining the positive effect of built-up extent in certain wards.

2. The presence of humans, not illumination or structural density, is the most critical factor in the suppression of vegetation, aligned with known patterns of resource extraction and intensification of land-use.
3. The functioning of the ecosystem is highly sensitive to demographic factors, suggesting that managing population-driven land conversion is essential to sustain green infrastructure in rapidly growing towns.

The findings show that the overall performance of the model ($R^2 = 0.862$) reveal that 86% of NDVI variability is accounting for by the three urbanization indicators, which underscore the central role of the urbanization dynamics in shaping the overall ecological conditions in the study locations. And also, the strong negative effect of the population density combined with the most modest positive effect of the built-up surfaces suggests that ecological degradation is predominantly linked to demographic expansion rather than structural development alone.

These findings reinforce the broader conclusion that vegetation productivity declines most sharply in high-density urban cores, where land conversion, resource pressure, and habitat fragmentation are most severe. In contrast, peri-urban areas with moderate built-up development but lower population densities may support more resilient vegetative cover.

3.6 Integrated Assessment of the impacts of urbanization

3.6.1 Synthesis of Findings between Indicators

Integration of LULC dynamics, plant diversity patterns, soil properties, vegetation indices, and urbanization metrics demonstrates that urbanization in Makurdi and Otukpo has produced a consistent trajectory of ecological alteration across the rural–peri-urban–urban gradient. Each data set highlights a distinct ecological dimension, yet together they reveal a unified pattern of plant community disruption and declining ecosystem functioning.

The LULC findings show that the built-up area expanded dramatically, increasing from 23.17 km² in 1990 to 143.11 km² in 2024, which represented 517% increase (119.94 km² gain) over the 34 years period of the study. Simultaneously, forest cover declined by 124.93 km², and cropland lost 159.71 km², indicating widespread conversion of natural and semi-natural habitats to urban surfaces. These quantitative shifts signify large-scale habitat loss, fragmentation, and replacement of

heterogeneous vegetated landscapes with ecologically simplified impervious environments. The findings also reveals that the Wetlands increased by 107.00 km², reflecting hydrological disturbance rather than ecological restoration, as their expansion coincided with vegetation removal and altered drainage, especially in rapidly urbanizing districts.

Biological field assessments confirm the ecological consequences of these changes. Furthermore, the findings of the analyses show that rural areas maintained the highest plant diversity, with Shannon index values averaging 2.83 in Makurdi and 2.74 in Otukpo, compared to lower values in peri-urban (2.49–2.54) and urban zones (2.64 in Makurdi; 2.52 in Otukpo).). Again somehow, it shows that the differences were not statistically significant, the biological gradient is unmistakable: increasing urban intensity corresponds to reduced species richness (23.50 → 15.83 species), lower ecological evenness and a shift toward disturbance-tolerant species. This pattern reflects ecological homogenisation, a well-documented biological effect of urbanisation where diverse plant communities are progressively replaced by generalists, ornamentals, and pioneer species.

Soil indicators further support the ecological shift. The urban soils of the findings show that the significantly higher pH values ($p = 0.037$) are greater than those of the peri-urban and rural soils, indicating increased alkalinity linked to anthropogenic inputs such as construction debris, wastewater infiltration, and reduced organic buffering. This finding further revealed that SOC and nitrogen have increased in urban zones (SOC increased from 166.27 to 211.99), but this pattern likely reflects episodic deposition of organic waste rather than improved nutrient cycling. True soil fertility is better reflected by the biological availability, structure, and organic turnover processes that are typically impaired in compacted and disturbed urban soils.

The vegetation indices showed the clearest ecological signal. NDVI declined from 0.482 ± 0.023 (rural) to 0.277 ± 0.076 (urban), while EVI decreased from 0.312 ± 0.025 to 0.173 ± 0.051 , demonstrating stark reductions in primary productivity and vegetation vigour. To nail it, the ANOVA results also confirmed highly significant differences between zones (NDVI: $F = 31.04$, $p < 0.001$; EVI: $F = 32.52$, $p < 0.001$), revealing that the functioning of vegetation decreases dramatically with increasing urban intensity.

The regression model synthesises these patterns by showing how specific components of urbanisation explain vegetation health. Then again shows that population density had the strongest negative effect on NDVI ($\beta = -0.0072$, $p < 0.001$), which identified demographic pressure as a key biological stressor driving the removal of vegetation, the extraction of fuelwood and the simplification of habitats. NTL, on the other hand, shows a very weak and a non-significant influence, thereby confirming that the economic intensity alone does not dictate vegetation status. The built extent showed a small but significant positive coefficient ($\beta = 0.0537$, $p = 0.020$), likely reflecting the mixed vegetative–settlement mosaics in the peri-urban areas rather than the true ecological benefit.

Collectively, the findings demonstrated that urban expansion (urbanisation) strongly affected biological ecosystems through multiple interacting pathways that caused habitat loss, soil modification, vegetation suppression, and demographic pressure converge to reshape the structure of the plant community and the functioning of the ecosystem.

3.6.2 Ecological Implications

The integrated evidence points to several key ecological consequences of urbanisation that directly impact plant diversity and ecosystem functioning in tropical cities such as Makurdi and Otukpo.

Habitat Fragmentation and Landscape Conversion

From the findings, the loss of 124.93 km² of forest and widespread decline in cropland have led to the fragmentation of formerly continuous habitats into isolated patches. Fragmentation reduces dispersal opportunities, increases edge effects, and disrupts plant community dynamics. This in turn further explained why the Forest-dependent species are particularly affected, leading to reduced niche diversity and potential local extinctions. Patch isolation also limits pollinator movement, seed flow, and successional processes essential for ecosystem recovery.

Biodiversity Decline and Ecological Homogenisation

The findings on the decline of the species richness from 23.50 species (rural) to 15.83 species (urban), along with reduced Shannon and Simpson indices, indicate that urbanisation fosters homogenisation of plant communities. Disturbance-tolerant

species such as *Cleome viscosa*, *Sida acuta*, and *Heterotis rotundifolia* become more dominant, while native forest species decline. This shift alone is considered to have reduced functional diversity, disrupted trophic and symbiotic relationships, and diminished ecological resilience.

Soil Fertility Modification and Biological Function Disruption

Significantly higher soil pH in urban areas ($p = 0.037$) indicates altered nutrient availability and potential micro-nutrient deficiencies that can suppress plant growth. Although SOC and nitrogen appear elevated, the biological reality is reduced nutrient turnover, poorer microbial activity, and compacted soil structure—conditions that limit root development and reduce soil–plant interactions fundamental to ecosystem processes.

Reduced Primary Productivity and Vegetation Health

The findings of NDVI, which clearly show the substantial decline in NDVI values ($0.48 \rightarrow 0.28$) and EVI ($0.31 \rightarrow 0.17$) demonstrate weakened photosynthetic activity, reduced biomass accumulation and lower canopy cover. This reduction compromises ecological functions such as carbon sequestration, microclimate regulation, soil stabilisation, and hydrological buffering. This confirms that the urban hotspots with the highest population densities show the lowest NDVI values, confirming that demographic stress is a major biological driver of ecosystem decline.

Mechanistic Insight from Urbanization Metrics

The regression results demonstrate that population pressure ($\beta = -0.0072$, $p < 0.001$) is the dominant suppressor of vegetation functioning. The significance is that high-density areas that typically experience more fuelwood harvesting, land clearing, waste dumping, and infrastructural expansion. The unexpected positive effect of the built intensity reflects the mixed peri-urban land cover rather than the true ecological improvement. These insights underscore that biological decline is primarily driven by human pressure, rather than structural development alone.

In general, urbanization in the Benue State is driving a clear transition from biologically rich, vegetated rural ecosystems to increasingly disturbed, low-diversity urban environments. The bigger picture is seen through the transformation that is marked by a loss of habitat complexity, reductions in plant species diversity,

alterations in soil chemistry and nutrient processes, suppressed vegetation productivity, and growing demographic pressure on ecological resources. And collectively, by extension these changes have mirrored broader patterns observed across tropical urban ecosystems, where the rapid and often unplanned expansion places intense stress on plant communities and disrupts the ecological processes that sustain them. The integrated findings presented by this study are therefore a strong biological basis for developing policy recommendations focused on biodiversity conservation, sustainable urban planning, and ecosystem restoration in rapidly growing tropical cities.

3.8 Discussion of the results

The results of this study provide a comprehensive understanding of how urbanization affects plant diversity and ecosystem functioning on rural–peri-urban–urban gradients in Makurdi and Otukpo. When examined alongside previous studies, the findings reflect the broader patterns observed in rapidly expanding urban cities in tropical regions and reinforce the established ecological theories on habitat modification, biodiversity decline, and ecosystem degradation.

3.8.1 Urban Expansion and Landscape Transformation

The high expansion of the built-up area across the study region for about 119.94 km² (517%) increased between the 1990 and 2024 period highlighted the scale and magnitude of the urban growth. The findings align well with other recorded global research that have demonstrated how rapid urban sprawl in African cities, where population increase and infrastructural expansion are major drivers of land-use change [[182]. The significant loss of forest cover (of about 124.93 km²) and cropland (of about 159.71 km²) have mirrored the patterns reported in other Nigerian cities such as Abuja, Lagos, and Port Harcourt, where built-up expansion has replaced vegetated [183].

Further implications can be seen by the observed increase in wetlands, especially after 2010, reflecting hydrological disturbance caused by land clearing, soil compaction, and surface runoff, a trend similarly noted in studies from Accra, Dar es Salaam, and Nairobi [117][118],<http://repository.out.ac.tz/id/eprint/4352>. These findings confirm that rapid and unplanned urbanization alters natural water flow patterns and exacerbates ecological imbalance. Overall, the LULC patterns in this

study correspond to established urban ecological theory which posits that urban growth creates structural simplification of landscapes, reduces habitat connectivity, and increases ecological fragmentation [186].

3.8.2 Plant Diversity Decline Along Urban Gradients

The reduction in plant species richness and diversity from rural to urban zones in the current study is consistent with those of previous findings that urbanization acts as a major ecological filter, favoring disturbance-tolerant and generalist species [187]. The rural areas of Makurdi and Otukpo exhibited the highest Shannon diversity (Makurdi: 2.83; Otukpo: 2.74), while the urban centers exhibited lower values, reflecting the simplified communities of plants. It is evident that the observed pattern is similar to those in tropical cities such as Kampala [188], Accra [157] and Ibadan [189], where the species richness declines sharply with increasing urban intensity.

However, the dominance of some species such as *Sida acuta*, *Cleome viscosa*, and *Heterotis rotundifolia* in urban plots is consistent with studies identifying these taxa as “hardy urban winners,” pioneers capable of surviving in disturbed soils, compacted environments, and polluted [61].). In contrast, the fact that there is a reduced presence of forest-dependent species mirrors the “urban loser” phenomenon described by [190], where habitat specialists decline as fragments of natural vegetation. Although ANOVA did not show significant statistical differences between zones, the ecological gradient remains biologically meaningful. Similar findings are reported in other tropical biodiversity studies, where variability within zones sometimes reduces statistical significance despite clear ecological patterns [191].

3.8.3 Soil Properties and Urban-Induced Alterations

The significant increase in soil pH in urban zones with a ($p = 0.037$) aligns closely with previous research which shows that the urban soils tend to be more alkaline as a result of the following causes; construction debris, waste deposition, and reduced organic buffering [192] The apparent increase in SOC and nitrogen in urban areas contradicts expectations of nutrient decline but is consistent with urban studies in West African cities where organic waste dumping temporarily raises SOC and N [119][120].). However, the implication of this is that such increases do not correspond to improved soil health; rather, they indicate the deposition of low-

quality organic material and reduced patterns of microbial activity widely reported in disturbed tropical urban settings [26][121].). These findings have clearly reinforced the concept that urbanisation can disrupt natural soil processes, which in turn can lead to a reduction of nutrient cycling, lower bioturbation, and altered soil structure [119]. The minimal variability in bulk density and CEC between zones may reflect the dominance of anthropogenic soil modification and mixed land-use impacts.

3.8.4 Decrease in vegetable productivity (NDVI/EVI)

The high reduction of the NDVI values in urbanisation zones (from 0.48 in rural to 0.28 in urban areas) and that of the EVI value range (from 0.31 to 0.17) indicates a decrease in the health and productivity of vegetation. The pattern of the declines is in agreement with the findings from other tropical cities where the NDVI values have also decreased significantly as the built-up intensity also grows, as well documented in the study of [196]. These declines reflect lower biomass, reduced canopy cover, and increased land surface modification, key symptoms of ecological stress in urban landscapes. However, significant ANOVA results ($p < 0.001$) have been closely aligned with other previous studies showing that vegetation indices are sensitive indicators of ecosystem functioning and respond strongly to anthropogenic disturbance [197].). The consistency of this trend in both Makurdi and Otukpo confirms that urbanization is a major determinant of the condition of vegetation in tropical settings.

3.8.5 Influence of Urbanization Metrics on Ecosystem Functioning

The findings of the regression model that show population density as the strongest negative predictor of NDVI ($\beta = -0.0072$, $p < 0.001$) reflect a well-established link between human population pressure and ecological [170] It is a fact that high-density areas often experience vegetation clearing, fuelwood extraction, infrastructure expansion, and unregulated land-use change factors also highlighted in studies from Lagos, Nairobi and Kumasi [122][123][124].).

On the other hand, the weak and non-significant influence of NTL corresponded to other findings that proved that the economic intensity alone does not necessarily determine ecological condition, but particularly in regions where informal settlements and peri-urban development fall outside formal electrification networks

[201]. Lastly and interestingly, the observed positive effect of the built-up expansion has mirrored other results from studies where mixed peri-urban landscapes maintain patches of vegetation within residential compounds, agroforestry systems, and undeveloped plots [202].). This reinforces the concept of “intermediate disturbance,” where moderate human activity can support mixed vegetation assemblages, although such areas remain ecologically vulnerable.

3.8.6 Synthesis with Tropical Urban Ecology Theory

Across all indicators considered in the current study, the findings strongly align with the broader conceptual framework of tropical urban ecology:

- Urbanization reduces plant diversity through habitat simplification and biotic homogenization.
- The functioning of the ecosystem declines as the productivity of vegetation decreases and the quality of the soil changes.
- The anthropogenic pressures especially the one coming from population density emerge as the strongest ecological drivers.
- Fragmentation and land conversion disrupt successional processes and ecosystem resilience.

These observed patterns and trends also matched global observations in which tropical cities are biodiversity hotspots under threat, where rapid, unplanned urban growth drives steep ecological transitions [203].

The combined evidence positions Makurdi and Otukpo within the broader narrative of tropical urban ecological change. In addition, the observed reductions, especially in the part of plant diversity, altered soil environments, and reduced vegetation productivity, reflect the cumulative impacts of land conversion, demographic pressure, and infrastructural expansion. Finally, the findings clearly aligned with the argument that the management of urban growth, the preservation of green spaces, and the restoration of degraded eco-systems are essential for maintaining the ecological stability in rapidly developing tropical regions.

CONCLUSION

The study has assessed the impacts of urbanization on plant diversity and ecosystem functioning across urbanization gradients in the Makurdi and Otukpo Local Government Areas of Benue State, Nigeria. By integrating land-use/land-cover analysis, field-based vegetation surveys, soil assessments, and statistical modelling, the study provides a comprehensive understanding of how rapid urban expansion alters ecological patterns in tropical urban environments.

The results revealed that the built-up areas increased by more than 517% between 1990 and 2024, replacing extensive forest and cropland. This large-scale conversion has also led to the following: habitat fragmentation, reduced ecological complexity, and altered hydrological patterns. Plant diversity exhibited a clear decreasing trend along the urban gradient, with rural areas maintaining the highest richness and Shannon diversity, while urban zones showed simplified plant communities dominated by disturbance-tolerant species. Also, the soil properties as revealed by the findings of the study, showed the urban-induced modifications, particularly a significant increase in soil pH, indicating disruption of natural nutrient processes. Vegetation indices (NDVI/EVI) further confirmed substantial declines in vegetation health and primary productivity toward urban Centre's. The regression modelling in its findings have demonstrated that population density was the strongest suppressor of vegetation greenness, highlighting demographic pressure as a major ecological driver. These findings collectively demonstrate that urbanization in Benue State is driving a transition from biologically rich rural ecosystems to ecologically degraded urban environments. The ecological consequences include biodiversity loss, reduced vegetation functioning, altered soil processes, and weakened ecosystem resilience.

Despite its strengths, this study has several limitations that should be acknowledged. First, is the temporal resolution of the remote sense data, which is constrained by the availability of cloud-free Landsat imagery for some years, which may have limited

the precision of finer-scale temporal analyses. Second, is the data and assessment of plant diversity that was conducted only in selected plots and did not incorporate seasonal variations, potentially influencing the detection of species that appear or decline at different times of the year. Third, there are soil data and analyses that focus primarily on key physicochemical indicators and do not include microbiological assessments as part of a whole, which are critical to fully understanding the ecological functioning of soil and nutrient cycle processes. Additionally, some urbanization indicators, such as built-up intensity and night-time light data, may not adequately capture the complexity of informal settlement patterns common in African cities. Finally, the regression modelling employed ward-level data aggregation, which, although useful for broad analysis, may obscure important micro-spatial variations occurring within individual neighbourhoods or land parcels.

To deepen our understanding of urban ecological dynamics in tropical cities, future research should adopt several methodological improvements. First, high-resolution satellite datasets such as Sentinel-2 and WorldView should be incorporated, and this is to enhance the precision of land-use/land-cover change detection. Multi-seasonal vegetation sampling is needed to capture temporal fluctuations in species composition and plant community structure. Secondly, the soil aspect of the study should be expanded to include micro-biome and enzyme activity analyses where lab test will be carried out, and will provide more detailed insights into ecosystem processes and soil health. Future work should also employ landscape fragmentation metrics such as edge density and patch cohesion to quantify ecological connectivity and habitat integrity. Lastly, urban ecological modelling tools such as InVEST and landscape metrics should be used to predict potential ecological outcomes in varying urban growth scenarios. In conclusion, investigating the socio-economic causes behind the loss of vegetation and land conversion would provide a more holistic and realistic understanding of urbanization processes and would support more integrated and effective planning strategies.

RECOMMENDATION

Based on the study findings, several actions are recommended to support and provide sustainable urban development and ecological conservation in Makurdi, Otukpo and similar tropical cities.

Strengthening Urban Planning and Land-Use Regulation

Enhancing urban sustainability in Benue State requires stronger planning frameworks and more effective land-use regulation. To achieve this fit, stricter zoning policies should be implemented to limit the uncontrolled expansion into ecologically sensitive areas. By establishing urban growth boundaries, it will help prevent further loss of forest and agricultural land, while the integration of ecological and geospatial data into development plans would ensure and provide more evidence-based decision-making.

Conservation and Restoration of Green Spaces:

The act and manner of protecting and restoring natural vegetation is critical to maintaining ecological integrity within rapidly expanding cities. The remaining forest fragments, riparian buffers, and other natural areas should be protected legally to preserve biodiversity hot spots. Reforestation and afforestation programmes in degraded peri-urban and urban zones can help recover lost habitats. In addition, establishing urban greenbelts and ecological corridors would improve habitat connectivity and support species movement across the landscape.

Sustainable Urban Landscaping and Biodiversity Enhancement:

It is recommended to improve urban landscaping practices to prioritise ecological sustainability and biodiversity conservation. The act and use of native plant species in landscaping should be encouraged to maintain local biodiversity and ecosystem processes. This fit can be achieved through the following; community-led initiatives, such as tree-planting campaigns, school greening programmes, and urban gardens, can play an important role in improving vegetation cover. Furthermore, regulating indiscriminate vegetation clearing during construction is crucial to reducing avoidable biodiversity loss.

Soil Quality Management:

Maintenance of healthy urban soils is essential to support vegetation growth and urban ecosystem services. It can be achieved through proper soil monitoring programmes that should be established to track changes in pH, organic matter, and nutrient status in different land-use zones. However, practices such as composting,

mulching, and the application of organic amendments can improve soil structure and fertility. Policies must also be enforced to minimise soil contamination arising from improper waste disposal and construction activities.

Climate and Ecosystem Resilience Strategies:

Urban resilience should be encouraged and improved by integrating climate responsive and ecosystem-based strategies into city planning. Expanding urban tree cover helps regulate the microclimate, mitigate the effects of heat-islands, and improve air quality. Natural-based solutions such as green roofs, permeable pavements, and wetland restoration should be incorporated into urban infrastructure to manage stormwater and improve ecological functioning. In addition, promoting alternatives to the use of fuelwood can help reduce the pressure on remaining vegetation.

Community Engagement and Environmental Education:

To gainfully achieve long-term environmental sustainability, active community participation is required and increased public awareness. Educational campaigns should be developed to inform residents about the ecological impacts of urbanization and the importance of biodiversity conservation. Government should always empower local communities to participate in conservation planning, and environmental stewardship strengthens collective responsibility for ecological resources. Incorporating environmental education into school curricula will also help nurture future generations of sustainability-minded citizens.

Policy Implications

The findings of this study have significant policy implications for urban planning, biodiversity conservation, and environmental management in Benue State and similar rapidly growing tropical cities. The revealed 517% increase in built-up area expansion as observed between a 34 year period, the extensive loss of forest and cropland, declining species diversity, soil chemical alteration, and reduced vegetation productivity collectively demonstrate that current patterns of urbanization are ecologically unsustainable. These findings have highlight the urgent need for integrated land-use policies that recognize vegetation and biodiversity as critical urban infrastructure rather than expendable land resources. The high decline

in the plant diversity across the urbanization gradients have underscores the need for state-level policies that can prioritize the protection of natural habitats and enforce compliance with environmental regulations during urban expansion projects.

The strong negative influence of population density on vegetation productivity, as observed and revealed from the study, suggests that demographic pressures must be a central consideration in urban policy formulation. Planners should incorporate ecological carrying capacity assessments into development approvals to prevent overcrowding and reduce vegetation removal associated with informal settlement growth. It is also important to note that significant changes in soil pH and nutrient conditions also point to the need for policies that promote sustainable soil management, including restrictions on indiscriminate waste disposal, erosion control measures, and the incorporation of green infrastructure in housing and commercial developments. These measures would help maintain soil health, which is essential for sustaining urban vegetation, mitigating the effects of heat islands, and improving the resilience of the ecosystem.

Furthermore, the observed reduction in the NDVI and EVI values in urbanisation gradients (zones) has implied that vegetation cover is rapidly diminishing, thus compromising essential ecosystem services such as carbon sequestration, microclimate regulation, stormwater retention, and air purification. Therefore, policymakers should prioritise the expansion of urban green spaces, including parks, community forests, riparian buffers, and tree-lined streets. Another way to address this issue is to integrate nature-based solutions, such as green roofs, permeable pavements, and restored wetlands into urban infrastructure, which could help address the environmental consequences of rapid urbanisation while improving the aesthetic and ecological value of urban neighbourhoods.

However, on a broader strategic level, the results highlight the need for evidence-based urban planning that incorporates geospatial data, ecological indicators, and predictive modelling. There should be a synergy between urban planners and decision makers in Benue State, and they could adopt geospatial tools to monitor land-use trends, identify ecological hotspots requiring protection, and forecast the ecological impacts of proposed development. In addition, the study findings provide a more reliable scientific foundation for updating state and local environmental policies, including Environmental Impact Assessment (EIA) procedures, urban

forest strategies, and climate adaptation frameworks. An important aspect to ensure is the strengthening of institutional collaboration between environmental agencies, town planning authorities, and community stakeholders to ensure that ecological considerations remain central to development decisions.

In general, the ecological trends revealed in this study call for a holistic policy shift from reactive environmental management toward proactive urban governance that combines sustainability. Lastly, through the integration of the biodiversity conservation sector, the soil protection sector, and green infrastructure into urban planning, the Benue State can mitigate the ecological costs of urbanisation while promoting healthier, more resilient and more liveable urban environments.

Contribution to Knowledge

The study highlighted numerous and important contributions to the scientific understanding of urban ecology in tropical environments. First, it introduces an integrated assessment framework that combines remote sensing, field-based biodiversity surveys, soil analyses, and statistical modelling, offering a methodological blueprint for similar research in other tropical cities. The second part, is that it provides rare empirical evidence from a poorly studied region by delivering one of the few comprehensive ecological assessments of urbanization impacts in Benue State, thereby expanding the limited literature on West African urban ecosystems. The third part is that the study demonstrates clear and significant urban gradient effects, showing how plant diversity, soil properties, and vegetation productivity change across rural, peri-urban, and urban zones, and thus contributing to broader global insights on ecological transitions in developing-world cities. The fourth part is that the identification of population density as the strongest predictor of vegetation decline advances understanding of the demographic pressures that drive ecosystem degradation in tropical urban landscapes. Finally, the study offers ecological insights that is relevant for policy making as well as presenting biologically grounded recommendations for sustainable urban planning, biodiversity conservation, and ecosystem restoration, directly enriching applied urban ecology and environmental management.

List of publications on the topic of the thesis

Articles in journals included in the List of peer-reviewed scientific publications, in which the main scientific results of dissertations for the degree of Candidate of Sciences, for the degree of Doctor of Sciences should be published:

1. Shaibu, O., Omokaro, G.O., Onyamoche, E.P., Tafese, G.T., Nafula, Z.S., Niambe, OC, Nnoli, E.C. (2025). Assessment of the environmental impact of urbanisation in the Metropolis of Makurdi, Benue State, Nigeria. *International Journal of Environmental Impacts*, Vol. 8, No. 2, pp. 393-400. <https://doi.org/10.18280/ije.080218> (Scopus Q3 and WOS Q3)
2. Shaibu ochoche, Pinaev Vladimir Evgenievich., Emaikwu Patience Onyamoche,Kavhiza Nyasha.John, Opara Onyinyechi Grace, and Niambe Obed Kohol. (2025), A social approach to assessing the impact of urbanisation on natural resources: A case study of Makurdi-Benue State, Nigeria *International Research Journal* No6 (156). — URL: <https://research-journal.org/en/archive/6-156-2025-june/10.60797/IRJ.2025.156.82> <https://doi.org/10.60797/IRJ.2025.156.82> Issue: No. 6 (156), 2025 (VAK K1(SCOPUS Q2, WOS Q1)
3. Emmanuel Igwe, Jerry Akomaye Ikwen’ Shaibu Ochoche, Aloye Racheal Aniah’ Joseph Ikwun Agiopu Ogar.(2025) Roles of Artificial intelligence in predictive Agroecological Modelling and sustainable land management in response to climate viability in the Moscow Region. XXVth International Multidisciplinary Scientific GeoConference Surveying, Geology and Mining, Ecology and Management (SGEM) DOI 10.5593/sgem2025/2.1/s07.07 ISSN 1314-2704 ISBN 978-619-7603-89-7 Conference Proceedings awaiting publication Q2 SCOPUS
4. Shaibu Ochoche. (2025) Impacts of urbanisation on Plant Diversity in Otukpo, Benue State, Nigeria. Young Scientific and Practical Conference. Department of Foreign Languages of the Faculty of Philology, Institute of Ecology RUDN University. (RSCI)
5. Ochoche Shaibu, Obed Kohol Niambe, Godspower Oke Omokaro, Zipporah Simiyu Nafula. (2025). Socio-ecological consequences of urban growth in dutsen kura gwari, Niger state, Nigeria, “Международный научно-исследовательский

центр “Endless Light in Science” Conference proceeding, DOI 10.24412/2709-1201-2025-31-93-101 (VAK, Library.ru, google scholar)

6. Godspower Oke Omokaro, Ikioukenigha Michael, Paul Obokparo Ewubare , Zipporah Simiyu Nafula , Obed Kohol Niambe , Shaibu Ochoche, Abdulmajeed Allan Chikukula and Elohor-Oghene Amarie. (2025). Soil fertility management practices among smallholder farmers in Ughelli North, Delta State, Nigeria. (2025). Tropical Agriculture, 102(3), 358–367.
<https://journals.sta.uwi.edu/ojs/index.php/ta/article/view/9040>

(SCOPUS Q4 & WOS Q4)

7. Shaibu Ochoche, Emmanuel Igwe, Vladimir Evgenievich Pinaev Emaikwu Patience Onyamoche (2025). Spatiotemporal Analysis of Land Use and Land Cover Changes Driven by Urbanisation in Makurdi, Nigeria (1990 to 2024): Implications for Cropland Loss, Environmental Risk, and Sustainable Urban Development (SCOPUS Q3 WOS Q3) UNDER REVIEW

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LIST OF ABBREVIATIONS

Abbreviations

AIC
ANOVA
BD
BIC
BuiltUP
CEC
CSV
D
E
EPI
ETM+
EVI
F-statistics
GEE
GHG
GIS
GPPmax
GPS
HAC
H'
HSD
IDH
IRJ
Km²
LGA
LM
LULC
mm
N
NDVI
NIR
NPC
NPP
NTL
OA
OLI
PA
POP
p-value
Q–Q Plot
RF
R²
SDG

Full Meaning

Akaike Information Criterion
Analysis of Variance
Bulk Density
Bayesian Information Criterion
Build-up Density (Urbanisation Metric)
Cation Exchange Capacity
Comma-Separated Values
Simpson's Diversity Index
Pielou's Evenness
Environmental Studies and Protection Issues
Enhanced Thematic Mapper Plus (Landsat)
Enhanced Vegetation Index
Fisher's Test Statistic
Google Earth Engine
Greenhouse Gas
Geographic Information System
Maximum Gross Primary Productivity
Global Positioning System
Higher Attestation Commission
Shannon–Wiener Diversity Index
Honestly Significant Difference
Intermediate Disturbance Hypothesis
International Research Journal
Square Kilometre
Local Government Area
Lagrange Multiplier
Land Use / Land Cover
Millimetres
Total Nitrogen
Normalised Difference Vegetation Index
Near-Infrared
National Population Commission
Net Primary Productivity
Night-Time Light
Overall Accuracy
Operational Land Imager (Landsat 8)
Producer's Accuracy
Population Density
Probability Value
Quantile–Quantile Plot
Random Forest
Coefficient of Determination
Sustainable Development Goals

SOC	Soil Organic Carbon
SVM	Support Vector Machine
TM	Thematic Mapper (Landsat)
UA	User's Accuracy
UHI	Urban Heat Island
VIIRS	Visible Infrared Imaging Radiometer Suite
VIF	Variance Inflation Factor
W	Shapiro–Wilk Test Statistic
WOS	Web of Science

APPENDICES

Appendix 1



APPENDIX 1 GEE
SCRIPT FOR EXPORT

Appendix 2

Location	Year	LULC Class	Area (km ²)	Area (%)
Makurdi	1990	Built-up Area	93.56	11.22
Makurdi	1990	Forest Area	420.17	50.40
Makurdi	1990	Crop Land	260.60	31.26
Makurdi	1990	Water Body	24.50	2.94
Makurdi	1990	Wetland	34.78	4.17
Otukpo	1990	Built-up Area	188.09	14.00
Otukpo	1990	Forest Area	637.01	48.00
Otukpo	1990	Crop Land	307.82	23.00
Otukpo	1990	Water Body	111.92	8.00
Otukpo	1990	Wetland	86.80	7.00
Makurdi	2010	Built-up Area	116.48	13.97
Makurdi	2010	Forest Area	214.73	25.76
Makurdi	2010	Crop Land	391.05	46.91
Makurdi	2010	Water Body	19.53	2.34
Makurdi	2010	Wetland	91.81	11.01
Otukpo	2010	Built-up Area	258.06	19.38
Otukpo	2010	Forest Area	538.83	40.46

Otukpo	2010	Crop Land	276.31	20.75
Otukpo	2010	Water Body	90.32	6.78
Otukpo	2010	Wetland	168.12	12.63
Makurdi	2024	Built-up Area	213.50	25.61
Makurdi	2024	Forest Area	295.24	35.42
Makurdi	2024	Crop Land	156.23	18.74
Makurdi	2024	Water Body	26.86	3.22
Makurdi	2024	Wetland	141.78	17.01
Otukpo	2024	Built-up Area	444.44	36.09
Otukpo	2024	Forest Area	345.10	28.02
Otukpo	2024	Crop Land	256.24	20.81
Otukpo	2024	Water Body	63.60	5.16
Otukpo	2024	Wetland	122.24	9.93

Appendix 3



Field Work on Plants Species Diversity Data Collection

Appendix 4



APPENDIX 4
WARD_CLASSIFICAT

Appendix 5



APPENDIX 5
WARD_BUFFER_DAT

Appendix 6



APPENDIX 6 COLAB
CODE FOR 3.3.txt

Appendix 7



APPENDIX 7 COLAB
CODE FOR 3.4.txt

Appendix 8



APPENDIX 8 COLAB
CODE FOR 3.5.txt

Appendix 9



APPENDIX 9.docx

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Date: 29th April, 2025

TO WHOM IT MAY CONCERN

Subject: Confirmation of Data Provision for Academic Research

This is to certify that Shualbu Ochoche a researcher from Institute of Environmental Engineering, People's friendship University Russia named after practice Lumumba (RUDN University), MOSCOW, Russia, Was provided plant-related data covering Makurdi local government area by our organization (Joseph Sarwuan Tarka University, Makurdi - Benue state-Nigeria) on request. The data was officially released to support his ongoing academic research and analysis. It includes records and information on (Brief description of the types of data e.g: plant distribution, species list, ecologies zones, etc), and was shared solely for academic and research purposes. We acknowledge that data originated from the records and provided directly to the above named researcher.

For further inquiries and verification, please feel free to contact our office

Sincerely,

John N. Iyabo 29/4/2025

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23RD May 2025

Date:

To Whom It May Concern

This letter serves as an attestation that the Data Collection survey, conducted by Surv. Idoko Frederick from April 12th to April 17th, 2025 has been completed in accordance with the agreed-upon terms and conditions. The survey focused on Data Collection from selected areas in state, utilizing Hand Held Global Positioning System (GPS). We confirm that the data collected is accurate and reliable, and that the survey results are a true representation of the information gathered.

By signing this letter, we attest that all statements herein are true and accurate to the best of my knowledge, information, and belief.

Please feel free to contact me if you have any questions or require further clarification on any of the point.

Thank you for your time and consideration.

Sincerely,

Surv. Idoko Frederick Okechi
Surveying & Geomatics Unit
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(Email: FredOkechi14@gmail.com)